SQL Server 2005 Seminar
Architectural Overview

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Solid Quality Learning

Architectural Overview

Agenda

- Introduction
- Transaction Management
- High availability & scalability
- Security

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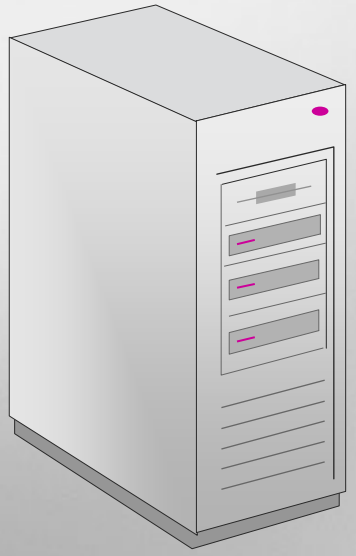
Introduction to SQL Server

Integrated services



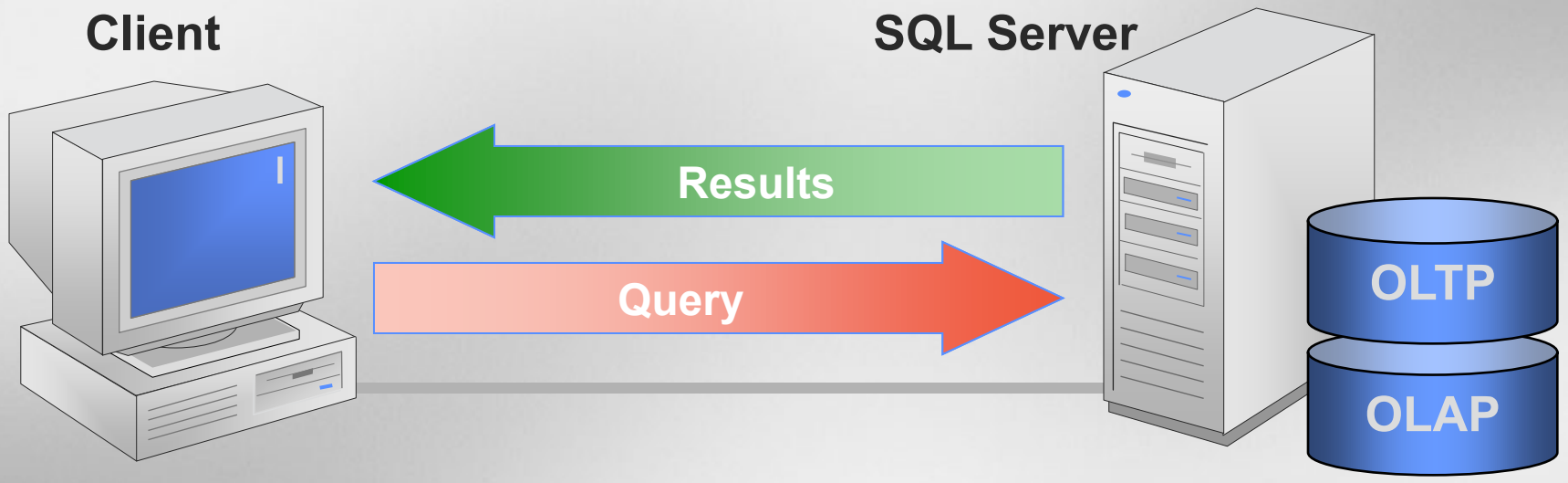
Server

- Database Engine
  - up to 50 instances
  - direct integration with MS-DTC
- SQL Server Agent
  - one for each instance of the database engine
- Full-text Search
  - one for each instance of the database engine
- Service Broker
- Integration Services
- Analysis Services
  - cluster aware, multiple instances
  - data mining
- Notification Services
- Reporting Services

- Both 32-bit and 64-bit versions

Introduction to SQL Server

Client-server model



Client Application

Relational Database Management System

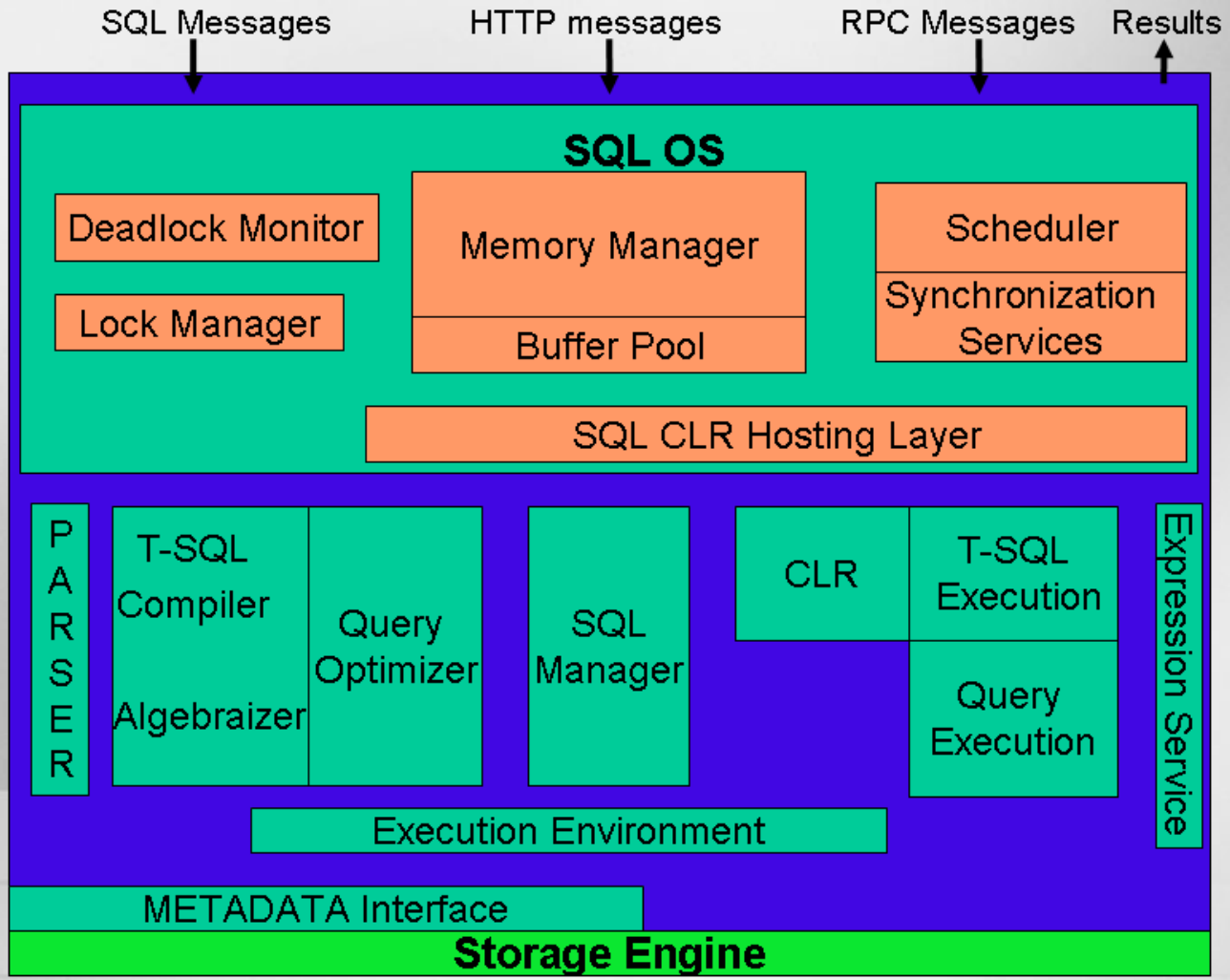
Introduction to SQL Server
Communications

- Based on the concept of Endpoints
  - an entry point for an instance

Transport Protocol	Payloads
Named pipes Shared Memory	TDS SOAP
TCP HTTP VIA	SSB Database Mirroring

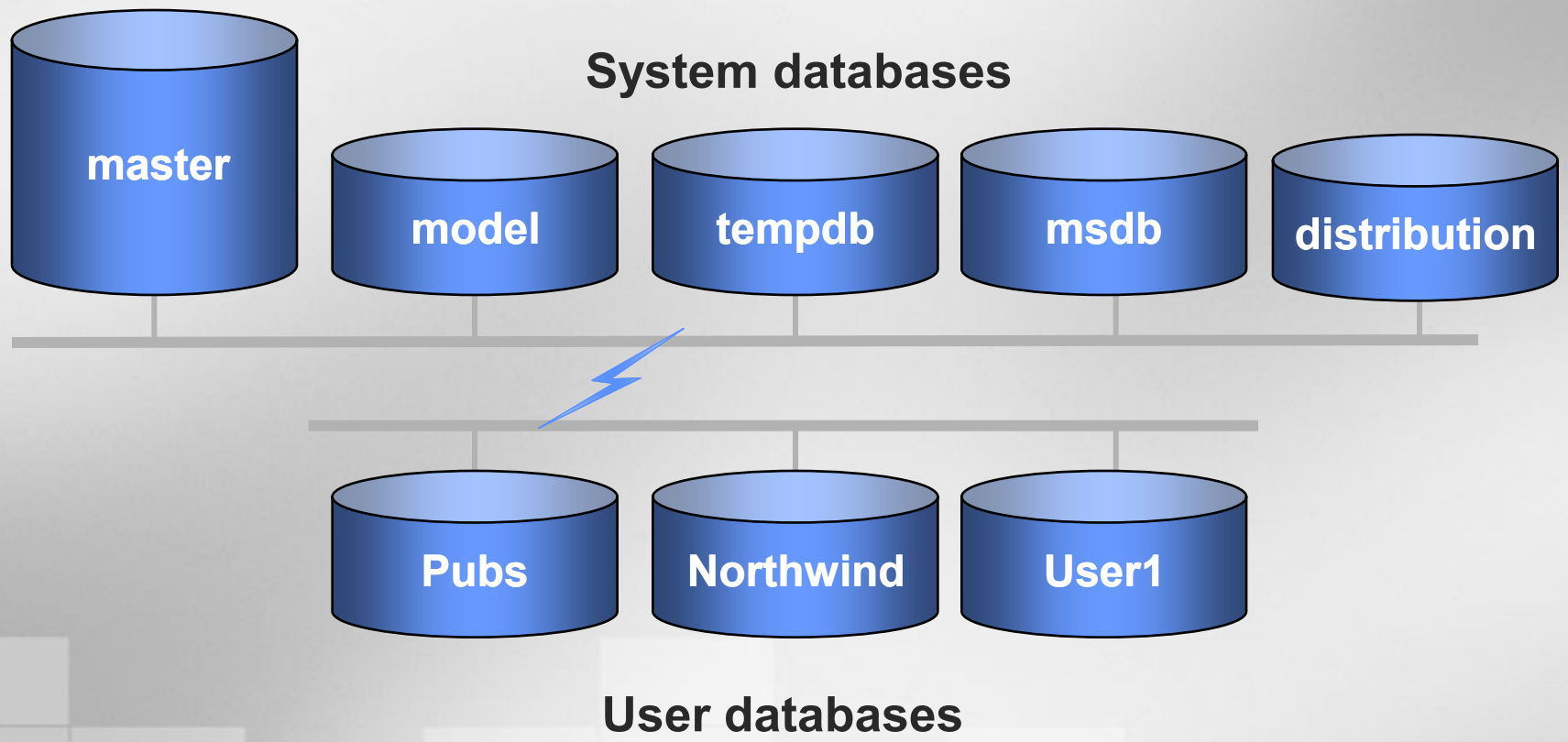
Introduction to SQL Server

Internal architecture overview



SQL Server Databases

Database types



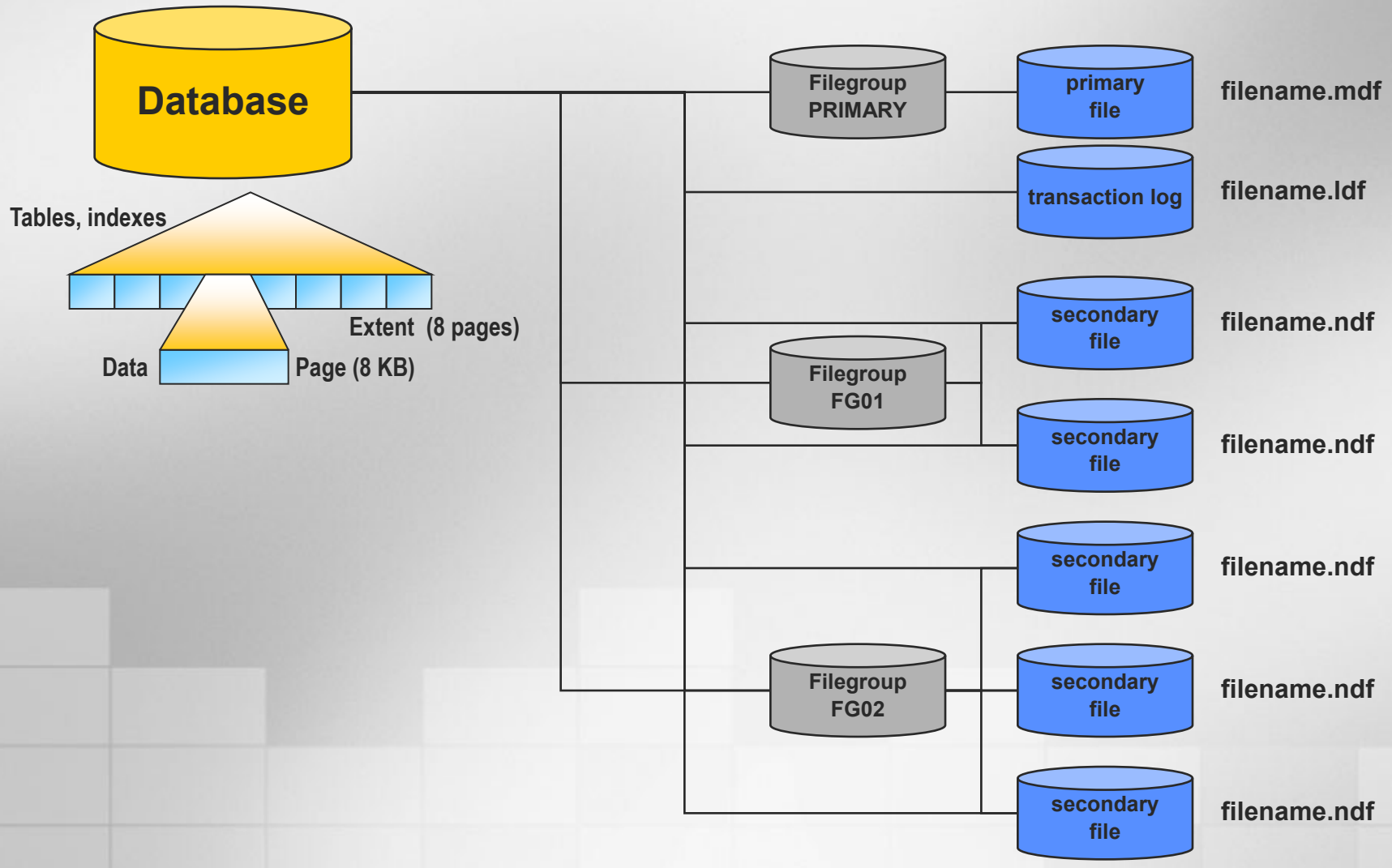
SQL server databases

Database objects

- Tables
  - columns names
  - columns data types
  - indexes
- Views
- Constraints
  - DEFAULT
  - CHECK
  - PRIMARY KEY
  - FOREIGN KEY
  - UNIQUE
  - Rules
- Stored procedures
- Triggers
- User defined functions
  - scalar
  - table valued
- User defined data types
- User defined aggregates
- Assemblies
- XML schemas

SQL Server databases

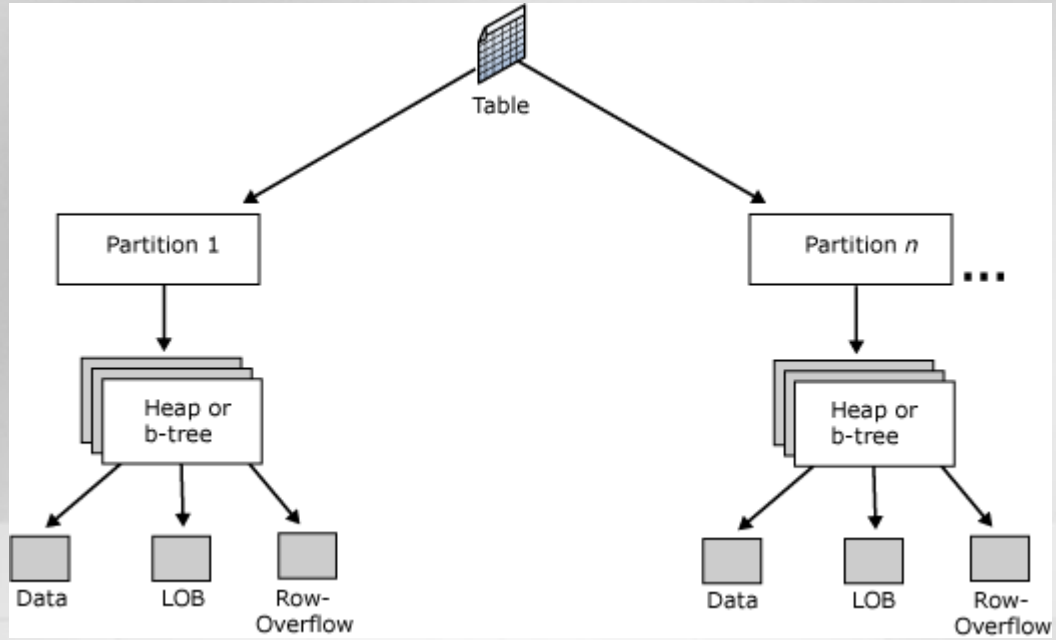
Physical structure



SQL Server databases

Physical implementation of tables

- Tables are implemented as
  - Heaps
  - B^+-Trees



SQL Server databases

Index types

- Primary Index (clustered)
  - leaf pages contain table data
- Secondary indexes
  - leaf pages contain RIDs or clustering keys
  - need to scan primary index with clustering key
  - online index reorganization and rebuild
- Specialized indexes for XML data type

Introduction to SQL Server

Instrumentation

- SQL Computer Manager
- Surface Area Configuration
- SQLCMD
- SQL Server Management Studio
- SQL Profiler
- Database Tuning Advisor
- SQL Server Business Intelligence Development Studio
- Visual Studio 2005

Introduction to SQL Server

Supported APIs

- OLE-DB
- ODBC
- SQL Native Provider
- JDBC
- db-library (unsupported)

- SMO
- WMI

Architectural Overview

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- High availability & scalability
- Security

Transaction management

In Transact-SQL

- Explicit transactions
  - `BEGIN TRANSACTION`
  - `COMMIT / ROLLBACK TRANSACTION`
- Implicit transactions
  - single commands
  - SET IMPLICIT_TRANSACTIONS ON
- Immediate checking transactions
  - automatic roll back depends on severity of errors

Transaction management

Nested transactions

- supported only syntactically
  - at most ONE transaction is open
  - commit of inner transactions are ignored
- **`BEGIN TRANSACTION`** increments **`@@trancount`**
- **`COMMIT TRANSACTION`** decrements **`@@trancount`**
  - when **`@@trancount`** reaches 0, the transaction **`COMMITs`**
- **`ROLLBACK`** sets **`@@trancount`** to 0
- Savepoints are supported
  - SAVE TRANSACTION
  - ROLLBACK to savepoint
- Useful to control transactions in Stored Procedures

Transactions isolation

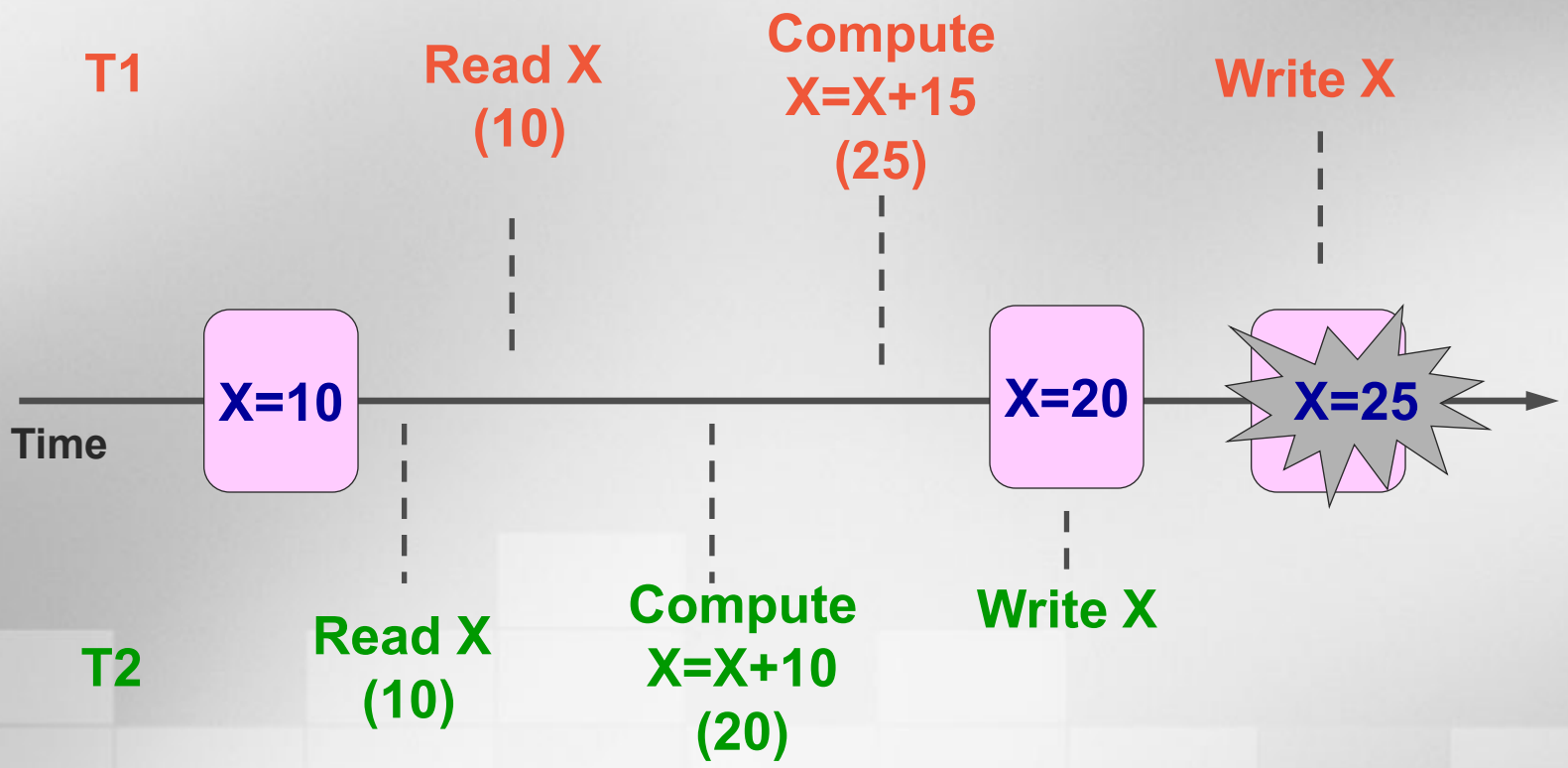
Common phenomenons

- Ideally all transactions should be serialized
  - maximum consistency
  - low concurrency
- If transactions are not completely isolated
  - Lost updates
  - Dirty reads
  - Non-repeatable reads
  - Phantom reads

Transaction isolation

Lost updates

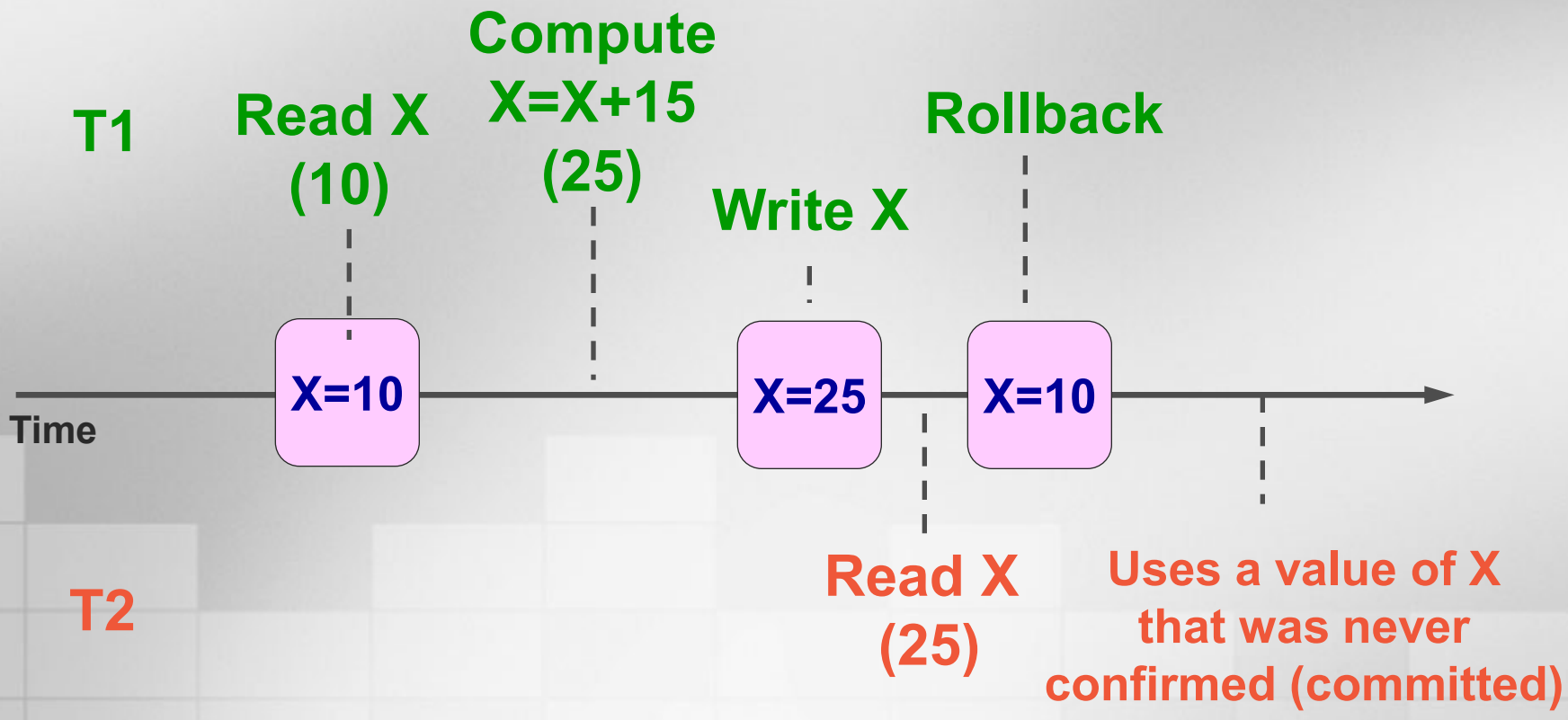
- Update not synchronized



Transaction Isolation

Dirty reads (inconsistent reads)

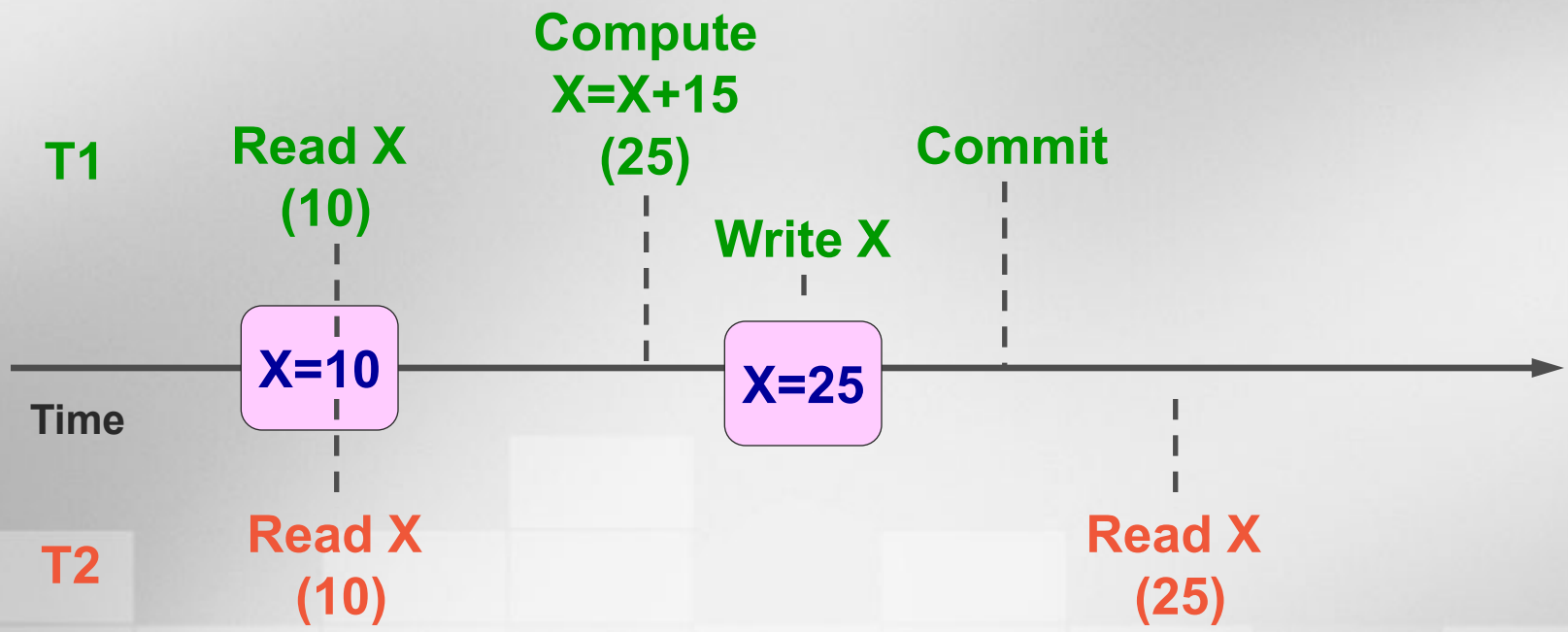
- T2 see uncommitted updates from T1



Transaction Isolation

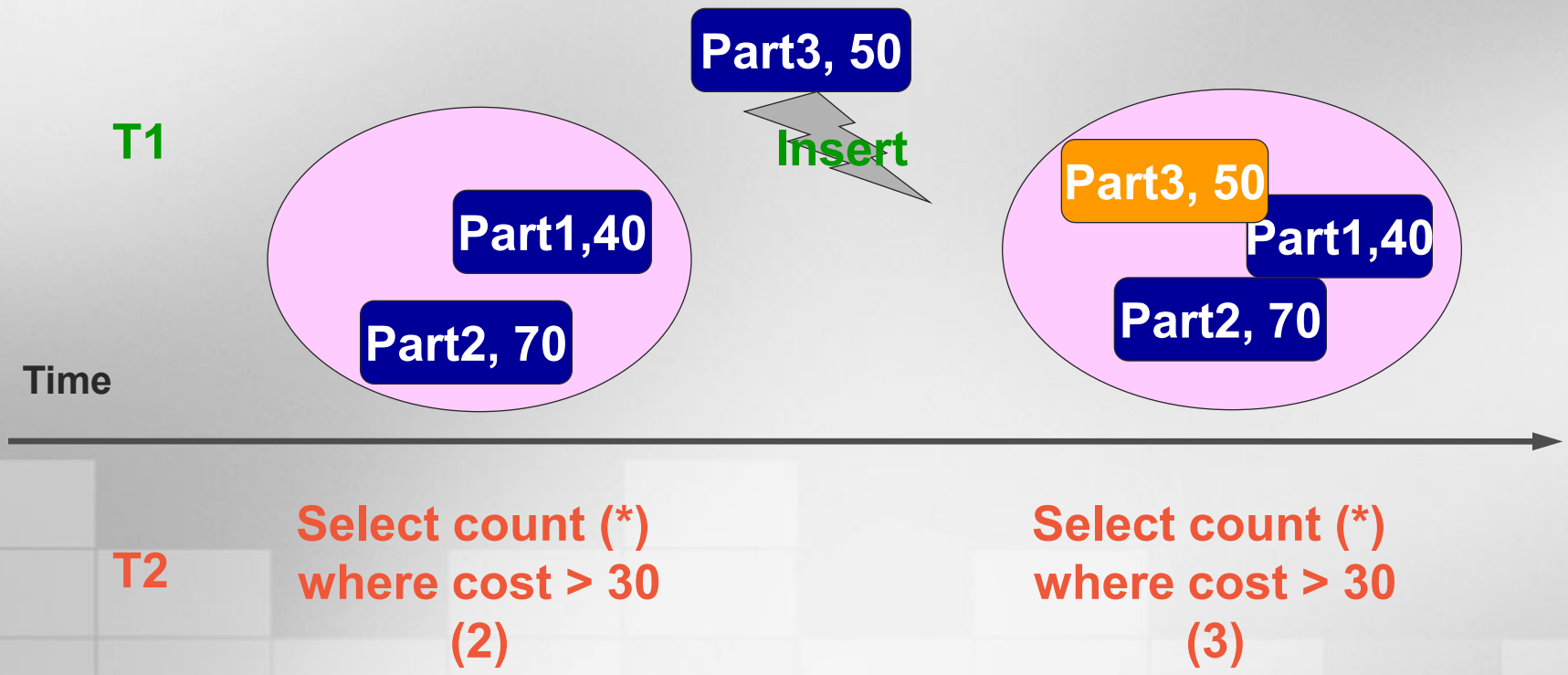
Non-repeatable reads

- Consistent read, but the data may change on subsequent reads



Transaction Isolation

Phantom reads



Transaction isolation
Support in SQL Server 2005

- Complete isolation is too costly
- ANSI SQL defines 4 isolation levels
  - serializable and 3 weaker levels
  - previous phenomenons are allowed to happen
  - it doesn’t specify the model to achieve isolation
    - optimistic
    - pessimistic
  - all previous SQL Server versions
    - based on pessimistic model
  - new version
    - implements an optimistic READ COMMITTED
    - implements a fifth optimistic level: SNAPSHOT

Transaction isolation

Levels, phenomenons and concurrency

Isolation level	Allowed phenomenon: Dirty Read	Non-Repeatable Read	Phantoms	Update Conflict	Concurrency model
Read Uncommitted	Yes	Yes	Yes	No	
Read Committed 1 Locking *2 Snapshot*	 No No	 Yes Yes	 Yes Yes	 No No	 **Pessimistic** **Optimistic**
Repeatable Read	No	No	Yes	No	**Pessimistic**
Snapshot	No	No	No	Yes	**Optimistic**
Serializable	No	No	No	No	**Pessimistic**

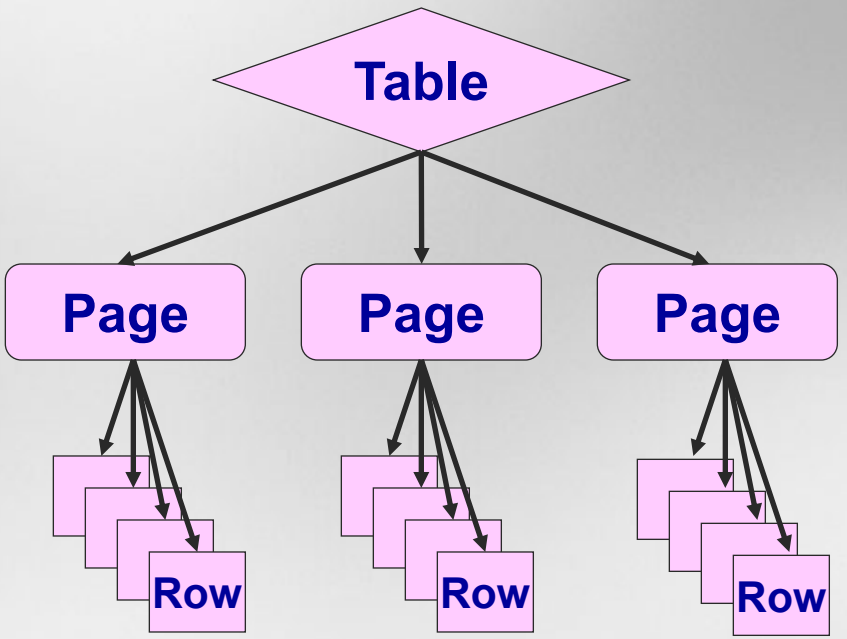
Transaction Isolation

Controlling the isolation level

- Session level option
- **`SET TRANSACTION ISOLATION LEVEL`**
  - Transact-SQL command for batches or stored procedures
  - APIs set the isolation levels under the covers
- Table Hints
  - isolation levels are tied to the lock manager behavior
  - hint in the SQL requests to force the lock manager
  - overrides current settings
- Balance
  - higher levels: more consistency, less performance
  - lower levels: less consistency, more performance
  - final decision based on specifications, not on performance!

Lock management
Overview

- Lock types
  - Shared (S)
  - Exclusive (X)
  - Update (U)
- Intent (I)
- Granularity
  - Row/key
  - Key range
  - Page
  - Table
  - Extent
  - Database



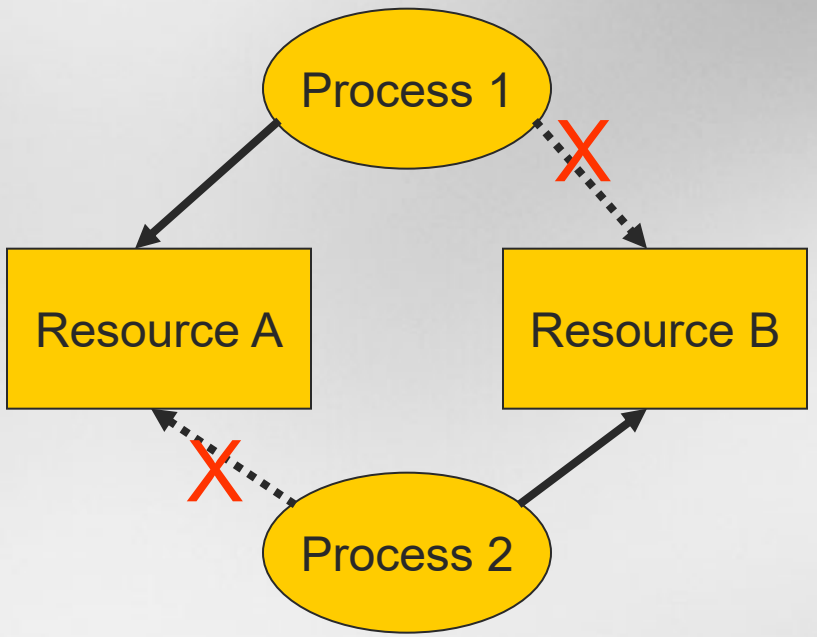
Lock management

Compatibility matrix

	Lock released							
Lock request	IS	S	U	IX	X	SchS	SchM	BU
IS	Yes	Yes	Yes	Yes	No	Yes	No	No
S	Yes	Yes	Yes	No	No	Yes	No	No
U	Yes	Yes	No	No	No	Yes	No	No
IX	Yes	No	No	Yes	No	Yes	No	No
X	No	No	No	No	No	Yes	No	No
SchS	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
SchM	No	No	No	No	No	No	No	No
BU	No	No	No	No	No	Yes	No	Yes

Lock management
Critical block (deadlock)

- Automatic resolution
  - checks for cycles regularly
  - transaction less costly to be canceled chosen as victim
  - transaction is rolled back
  - checks more frequently if deadlocks occurs

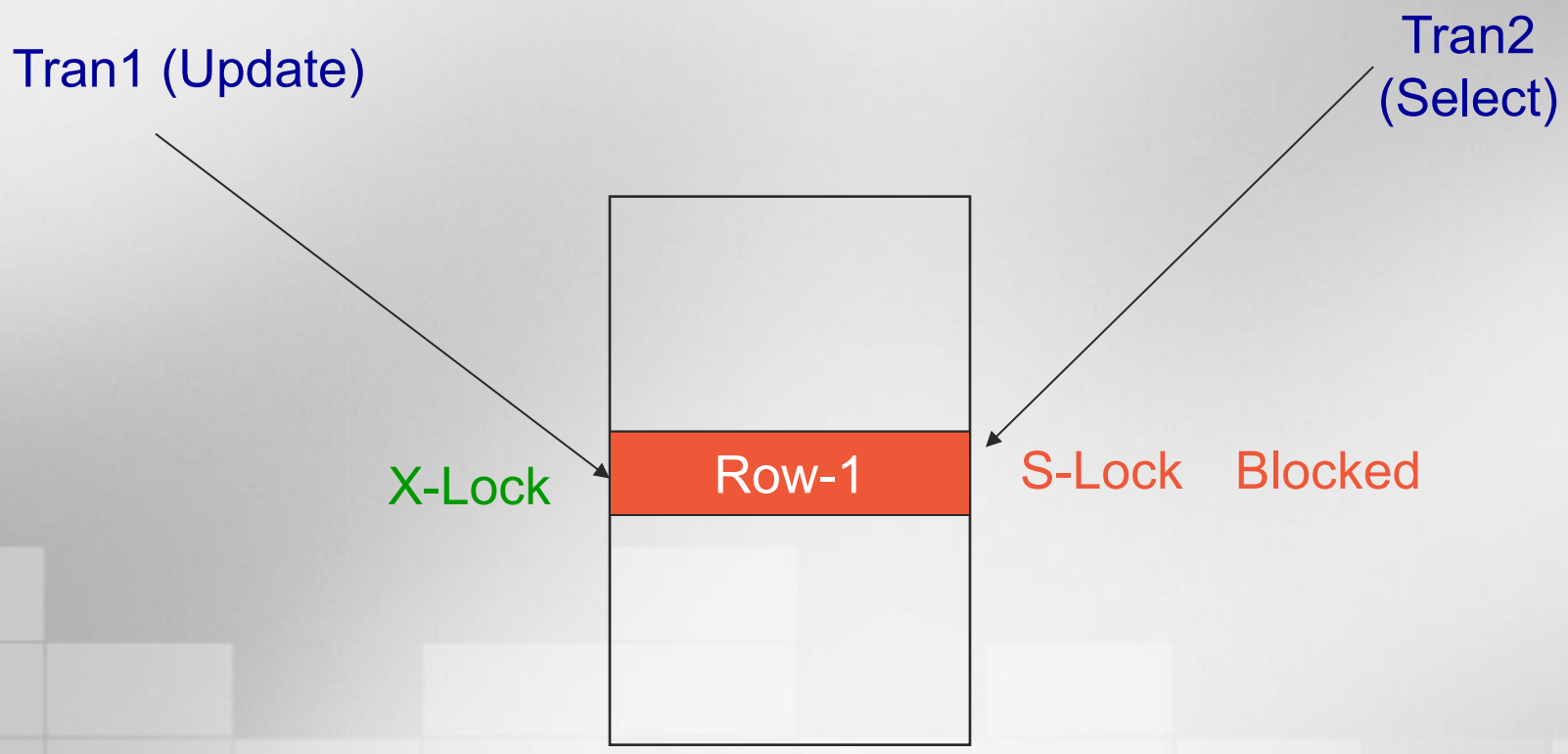


- Dealing with
  - error code
    - resubmit transaction
  - profiler event
    - full dump of lock graph

Transaction isolation

Problem with RC pessimistic model

- Write operation block a read operation



Transaction isolation

Read Committed Snapshot

```
CREATE TABLE t1 ( c1 int unique, c2 int)
INSERT INTO t1 VALUES (1, 5)
```

Transaction 1

```
BEGIN TRAN

UPDATE t1
SET    c2 = 9
WHERE  c1 = 1

COMMIT TRAN
```

Transaction 2
(Read Committed Snapshot)

```
BEGIN TRAN

SELECT c2 FROM t1
WHERE  c1 = 1

  -- SQL Server answers 5

SELECT c2 FROM t1
WHERE  c1 = 1
  -- SQL Server answers 9

COMMIT TRAN
```

Transaction isolation

Snapshot Isolation: read

```
CREATE TABLE t1 (c1 int unique, c2 int)
INSERT INTO t1 VALUES (1, 5)
```

Transaction 1

```
BEGIN TRAN
UPDATE t1
SET    c2 = 9
WHERE  c1 = 1


COMMIT TRAN
```

Transaction 2 (Snapshot Isolation)

```
SET TRANSACTION ISOLATION LEVEL
SNAPSHOT
BEGIN TRAN
SELECT c2 FROM t1 WHERE c1 = 1
  -- SQL Server answers 5
SELECT c2 FROM t1 WHERE c1 = 1
  -- SQL Server answers 5
COMMIT TRAN
SELECT c2 FROM t1 WHERE c1 = 1
  -- SQL Server answers 9
```

Transaction isolation

Snapshot Isolation: update

```
CREATE TABLE t1 ( c1 int unique, c2 int)
INSERT INTO t1 VALUES (1,5)
```

Transaction 2

```
BEGIN TRAN

UPDATE t1
SET    c2 = 9
WHERE  c1 = 1

COMMIT TRAN
```

Transaction 1 (Snapshot Isolation)

```
SET TRANSACTION ISOLATION LEVEL
SNAPSHOT

BEGIN TRAN

SELECT c2 FROM t1 WHERE c1 = 1

  -- SQL Server answer 5

UPDATE t1
SET    c2 = 15
WHERE  c1 = 1
```

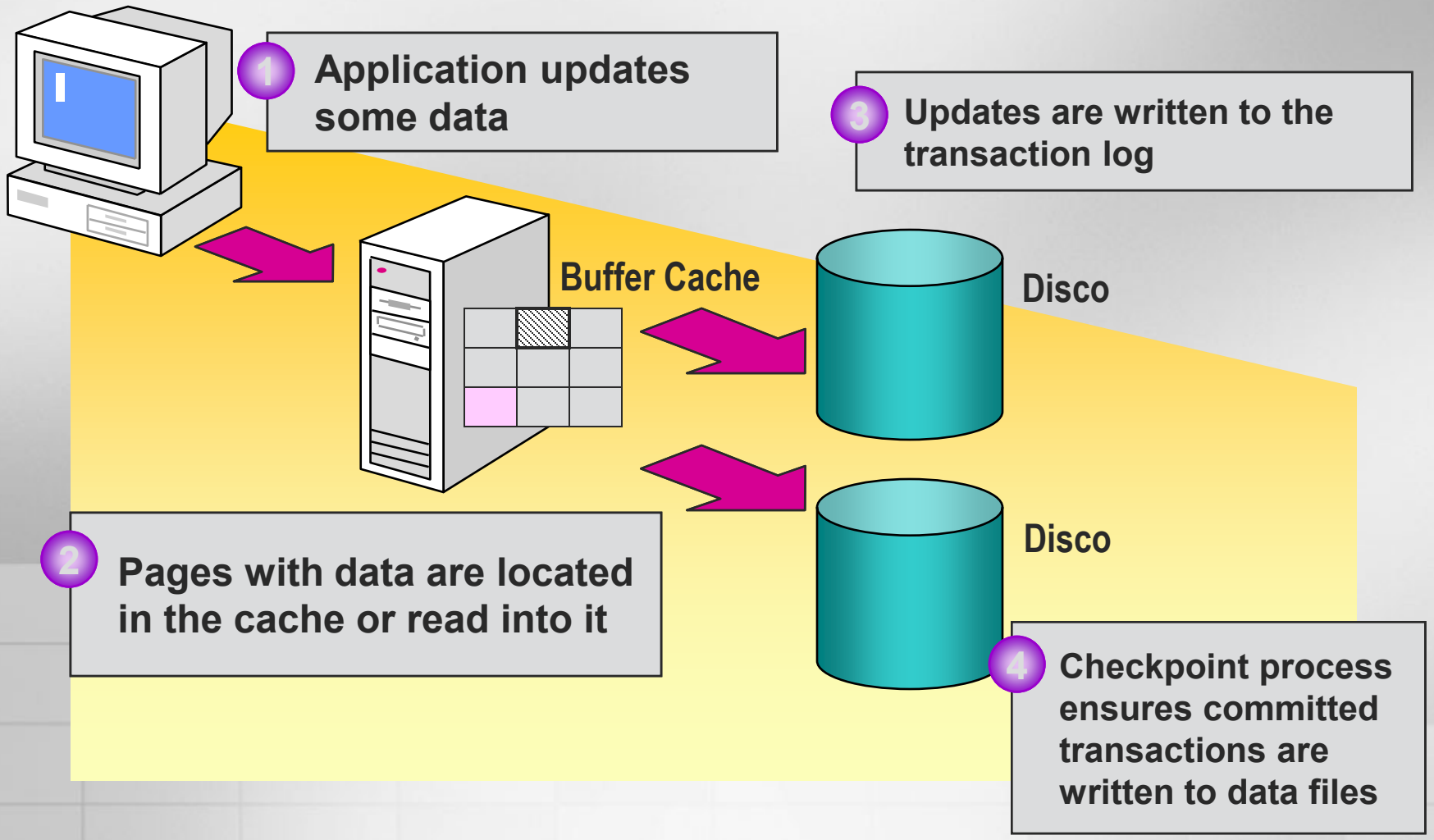
Blocked!

Time

Rollback because of a conflict with updates!

Transaction log

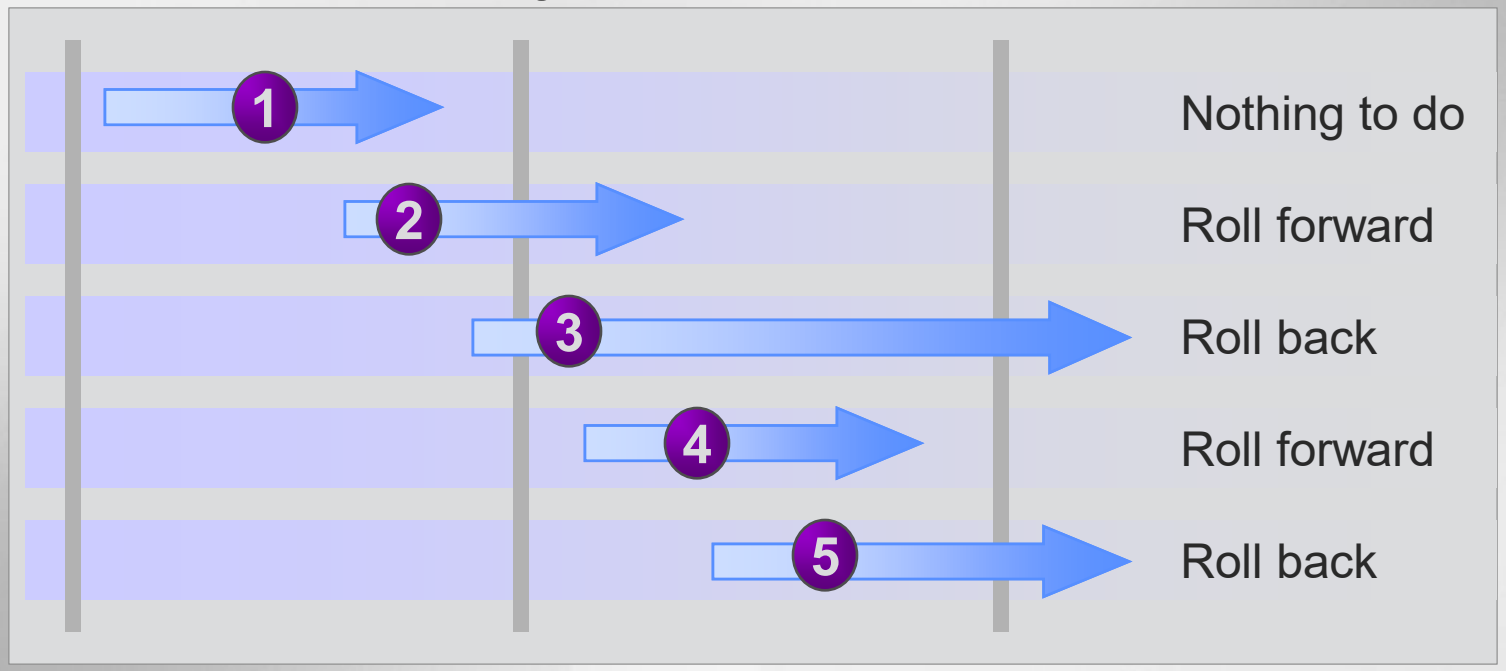
Overview: write ahead log



Transaction log

Restart (crash) recovery and checkpoint

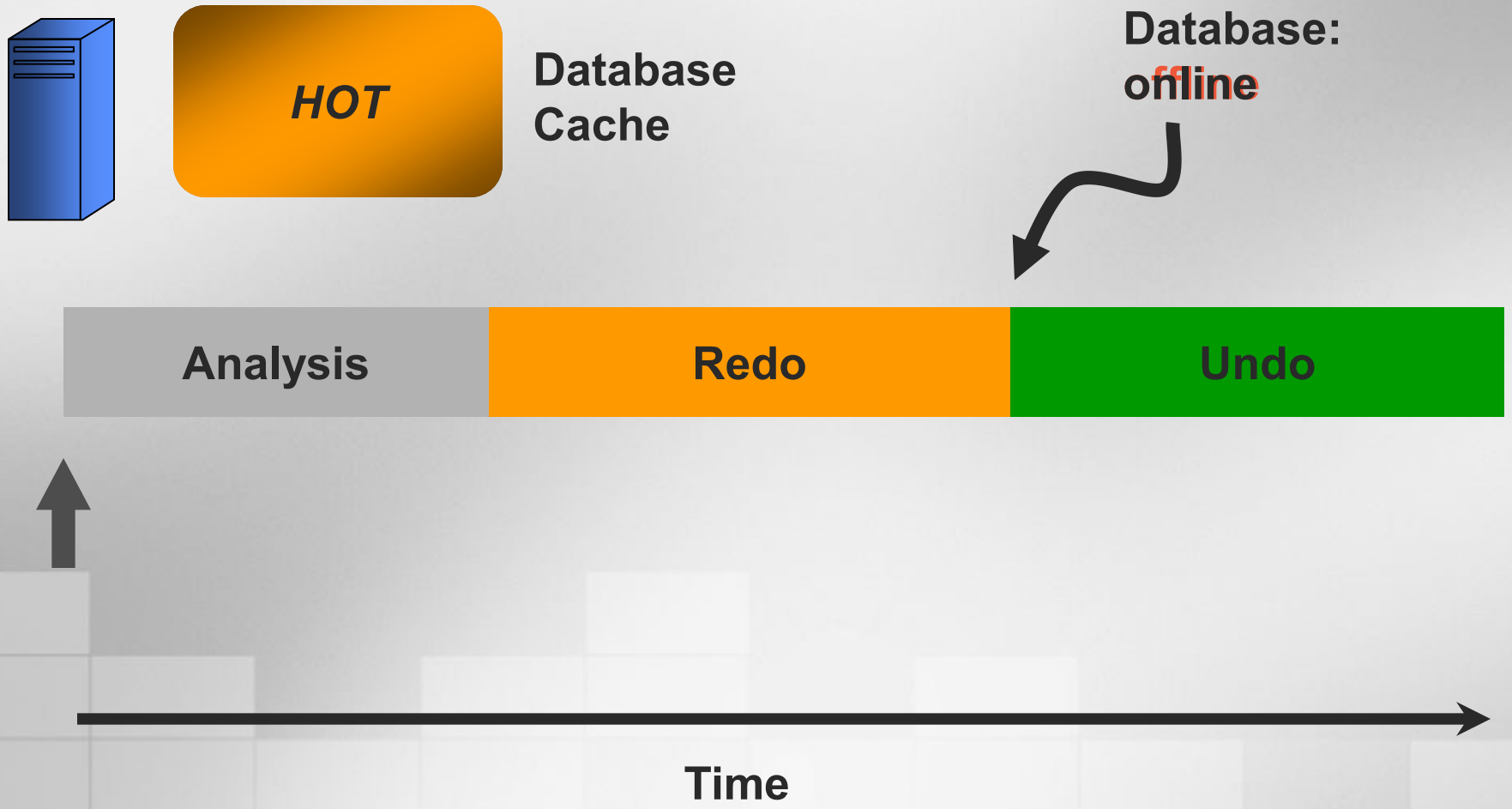
Transaction recovery



Checkpoint **Crash**

Transaction log

Optimized restart recovery



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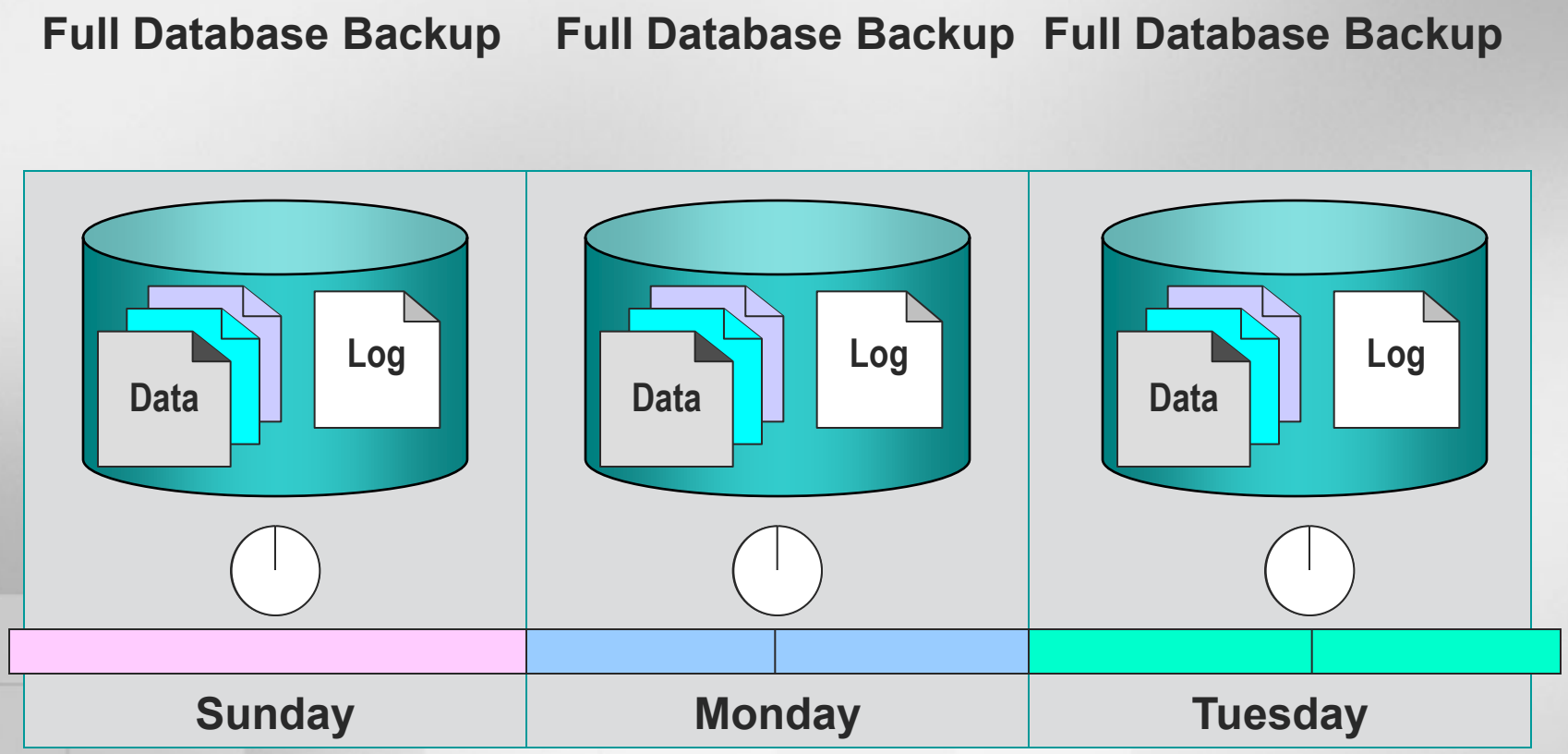
Media recovery (restore)

Introduction

- Backup
  - full
  - transaction log
  - differential
  - files and filegroups
  - consistent with the end time
- Restore
  - possible to do it online and fragmented
    - if database is split among filegroups
  - uses the same mechanism of crash recovery

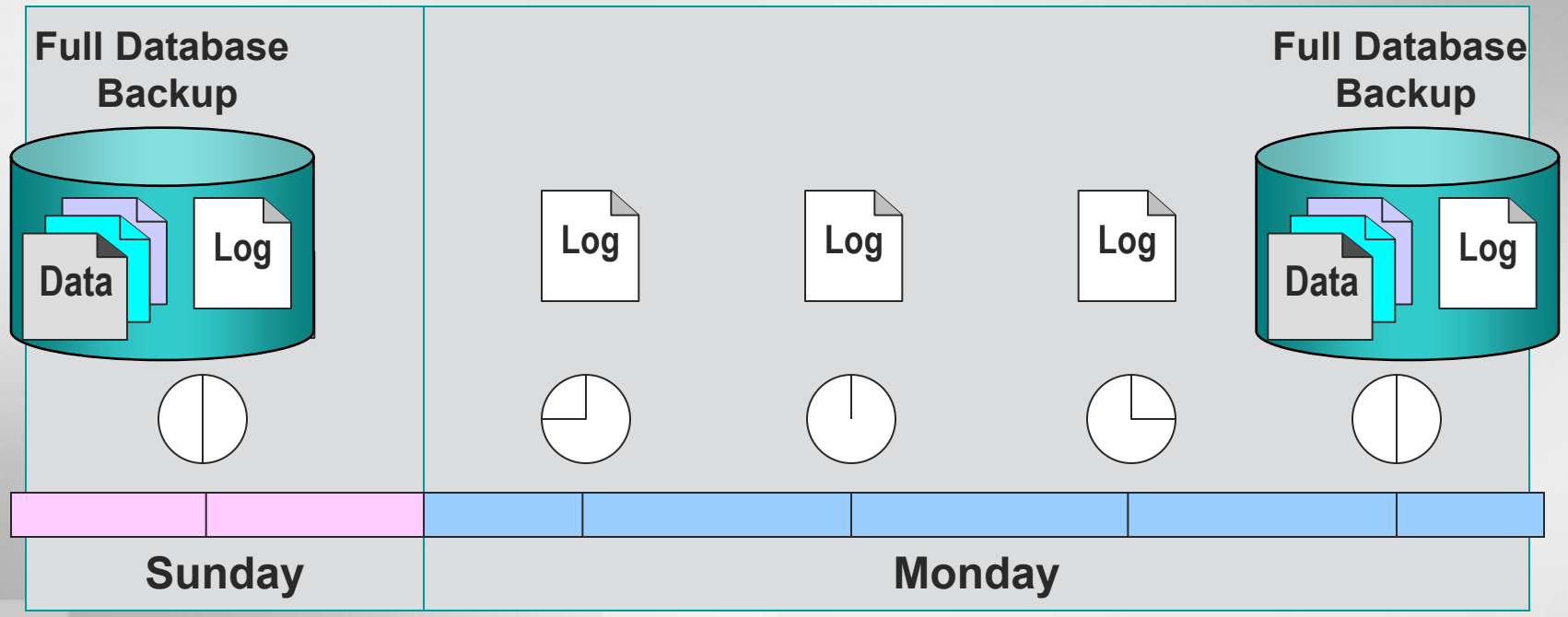
Media recovery

Example: full backup



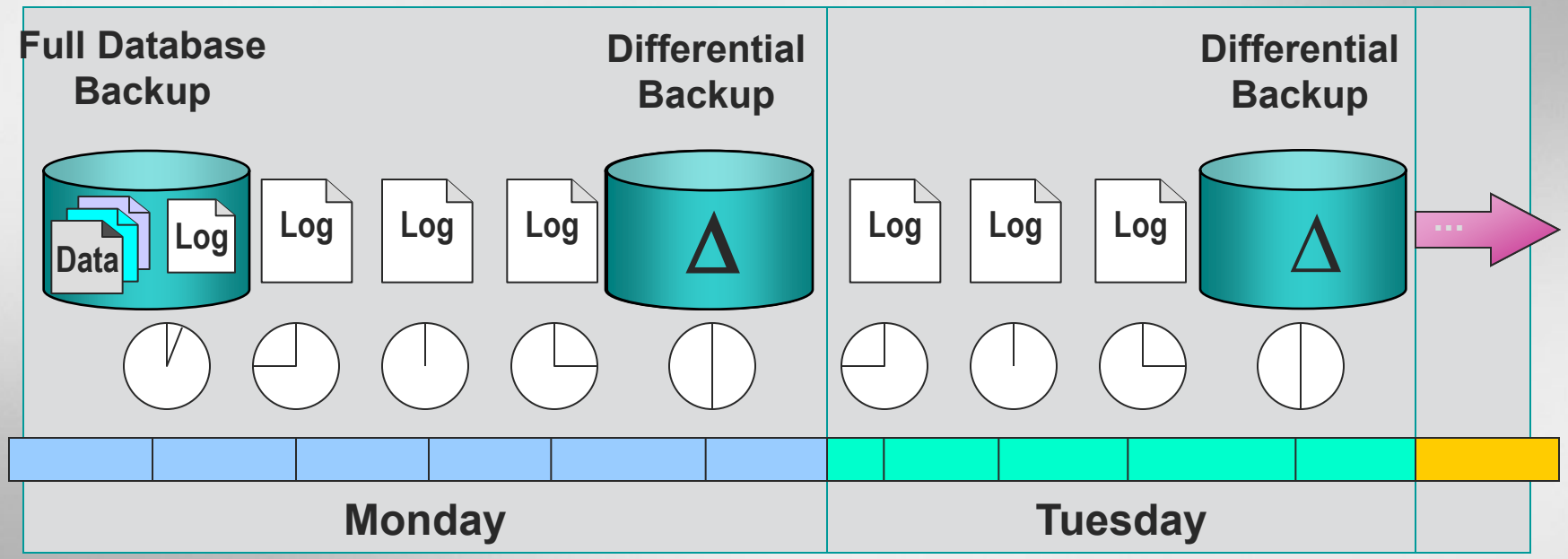
Media recovery

Example: full and log backup



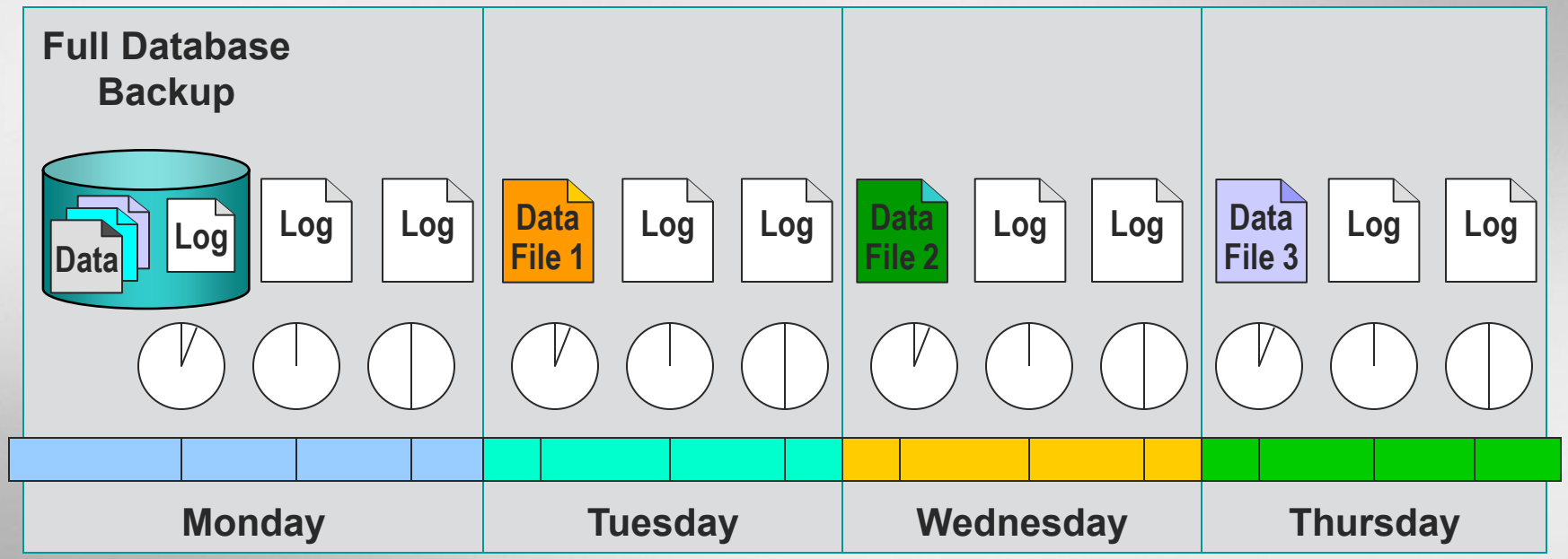
Media recovery

Example: differential and log backup



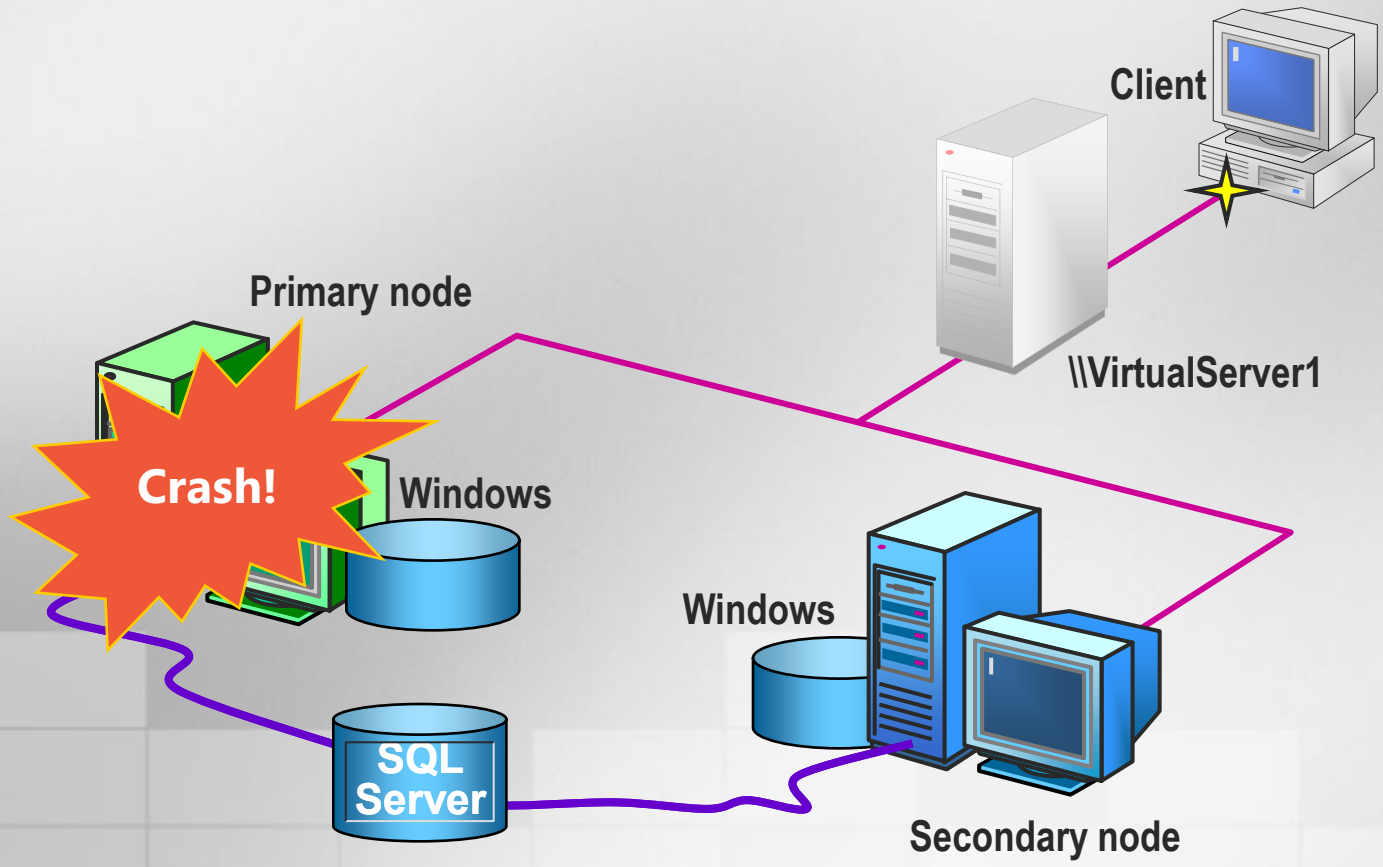
Media recovery

Example: files and filegroups



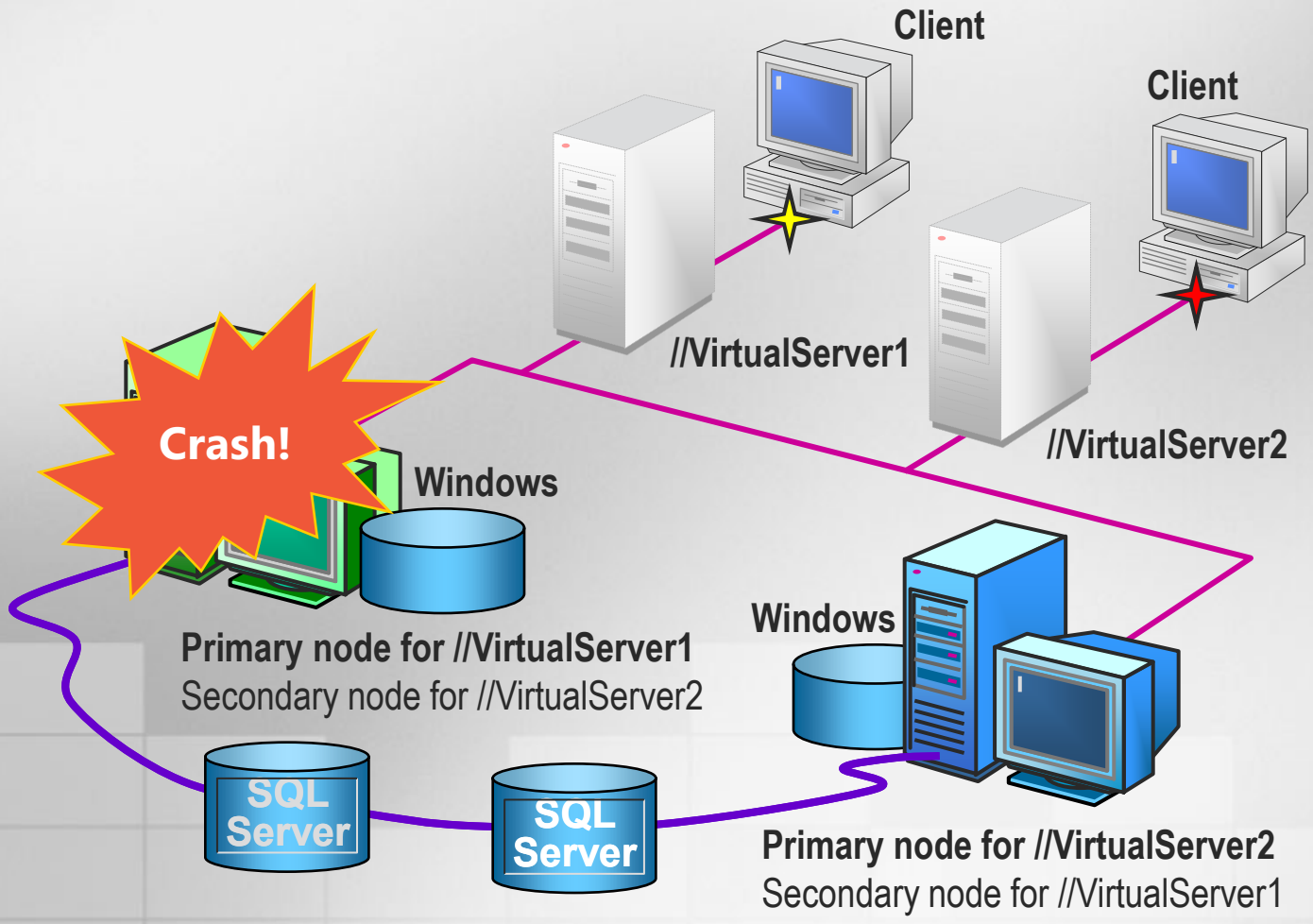
Failover Clustering

Active-Passive

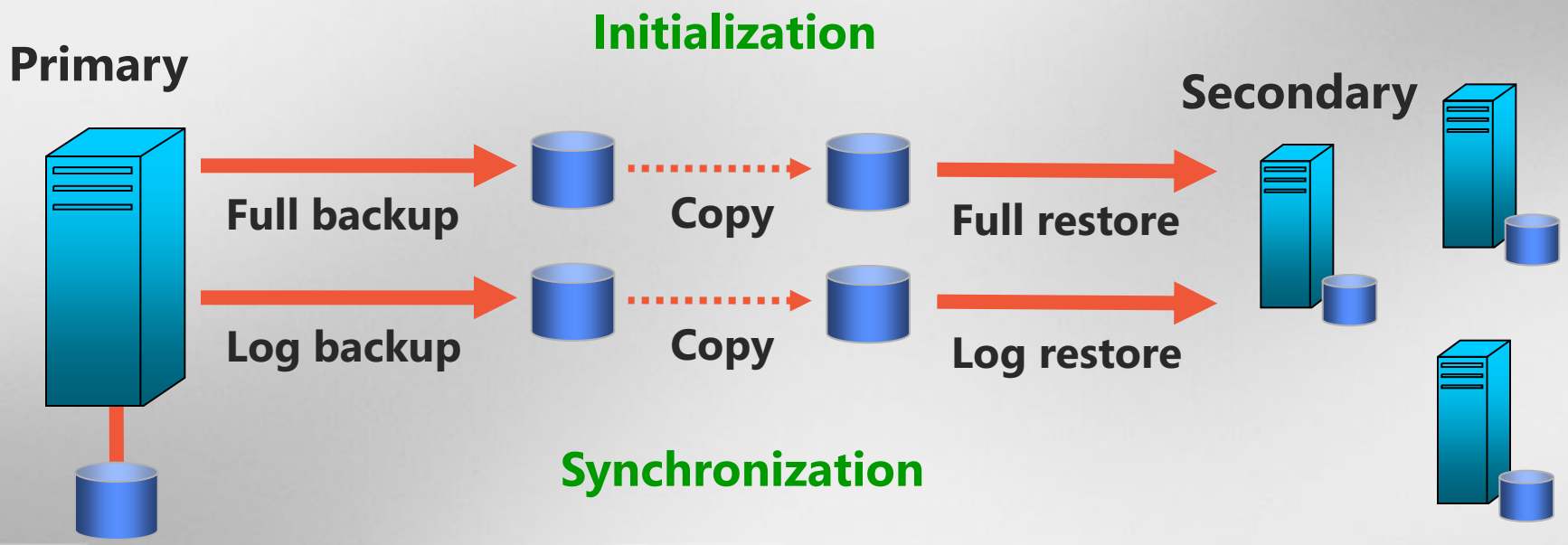


Failover Clustering

Active-Active

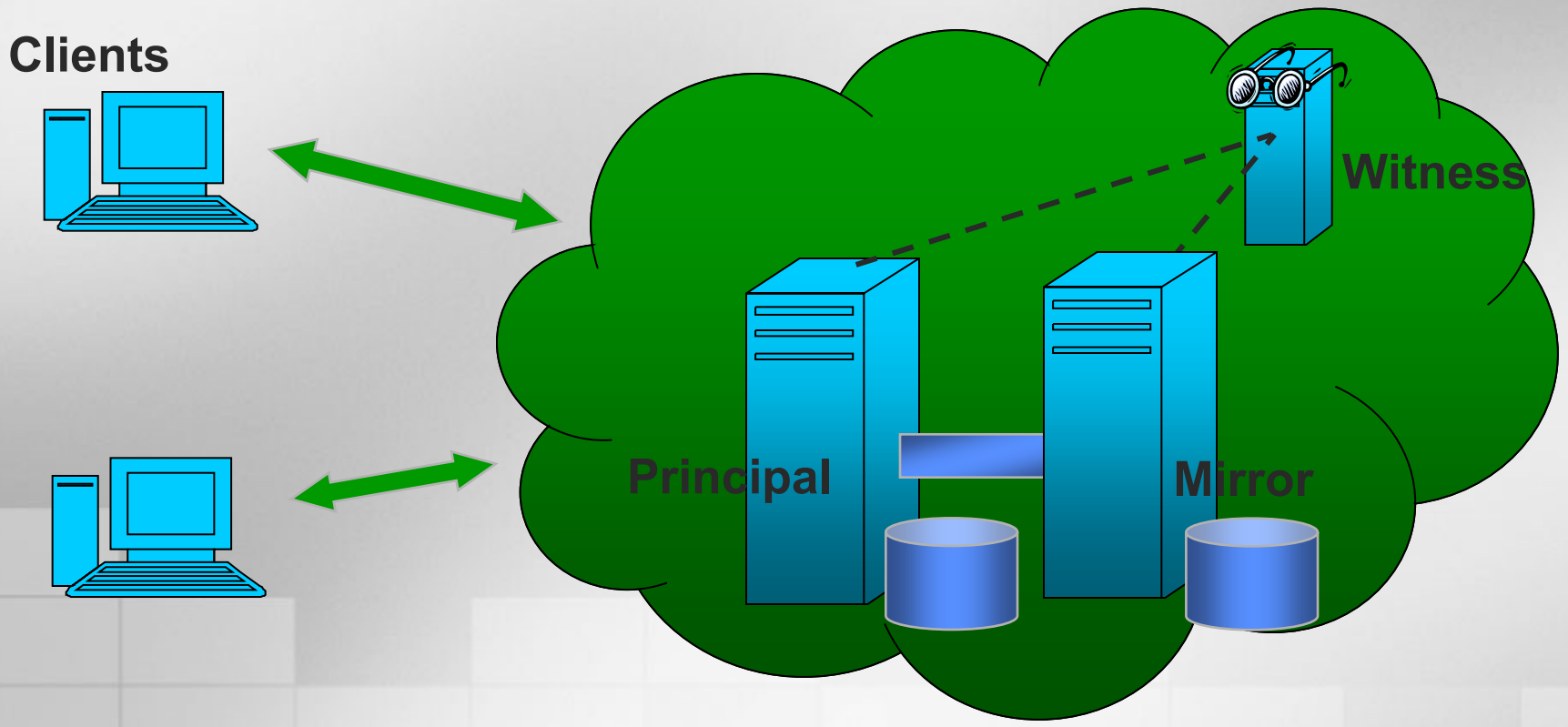


Log Shipping
Overview



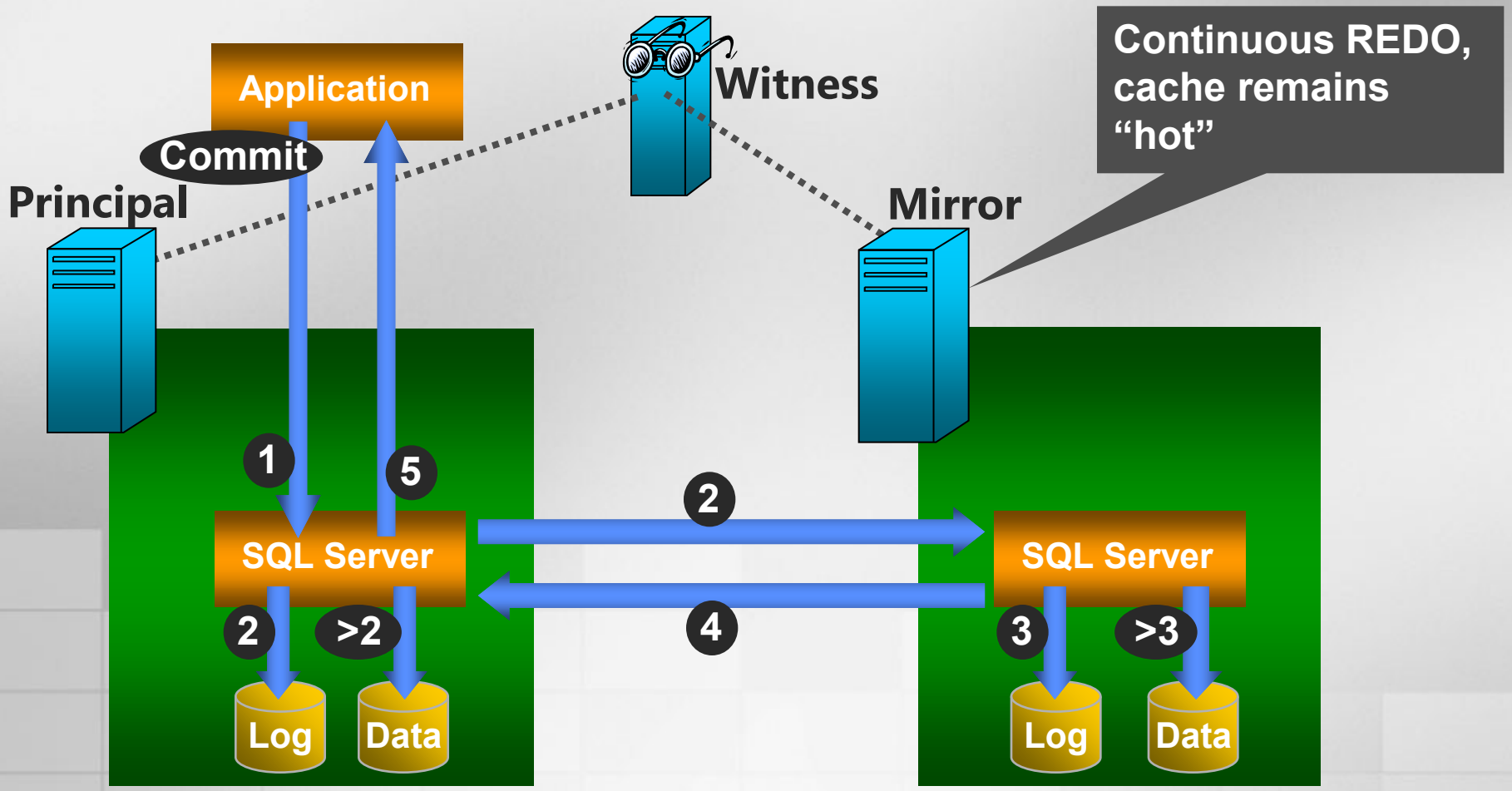
Database Mirroring

Overview



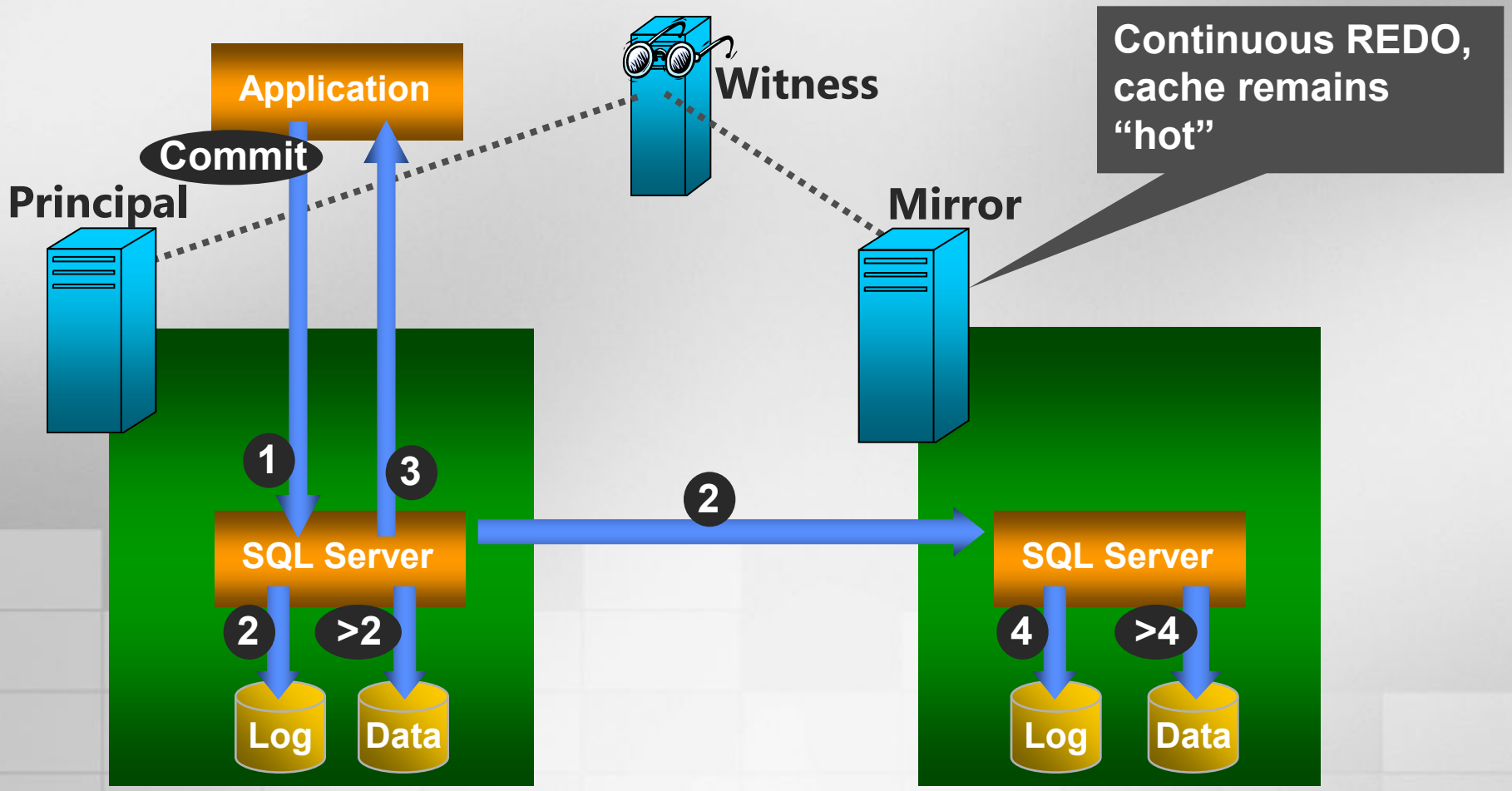
Database Mirroring

Safety FULL model



Database Mirroring

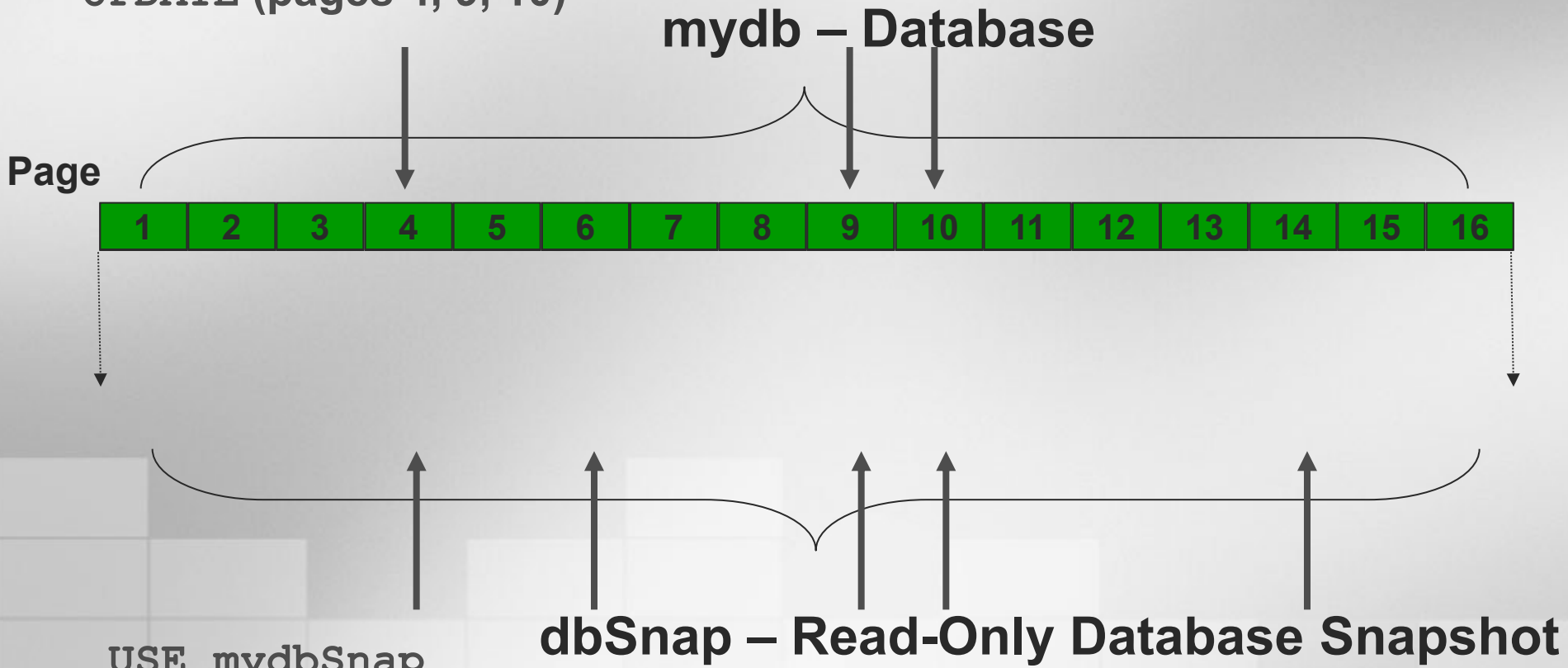
Safety OFF model



Database Snapshots

How does it work at the logical level

```
CREATE DATABASE dbSnap AS SNAPSHOT OF mydb
USE mydb
UPDATE (pages 4, 9, 10)
```

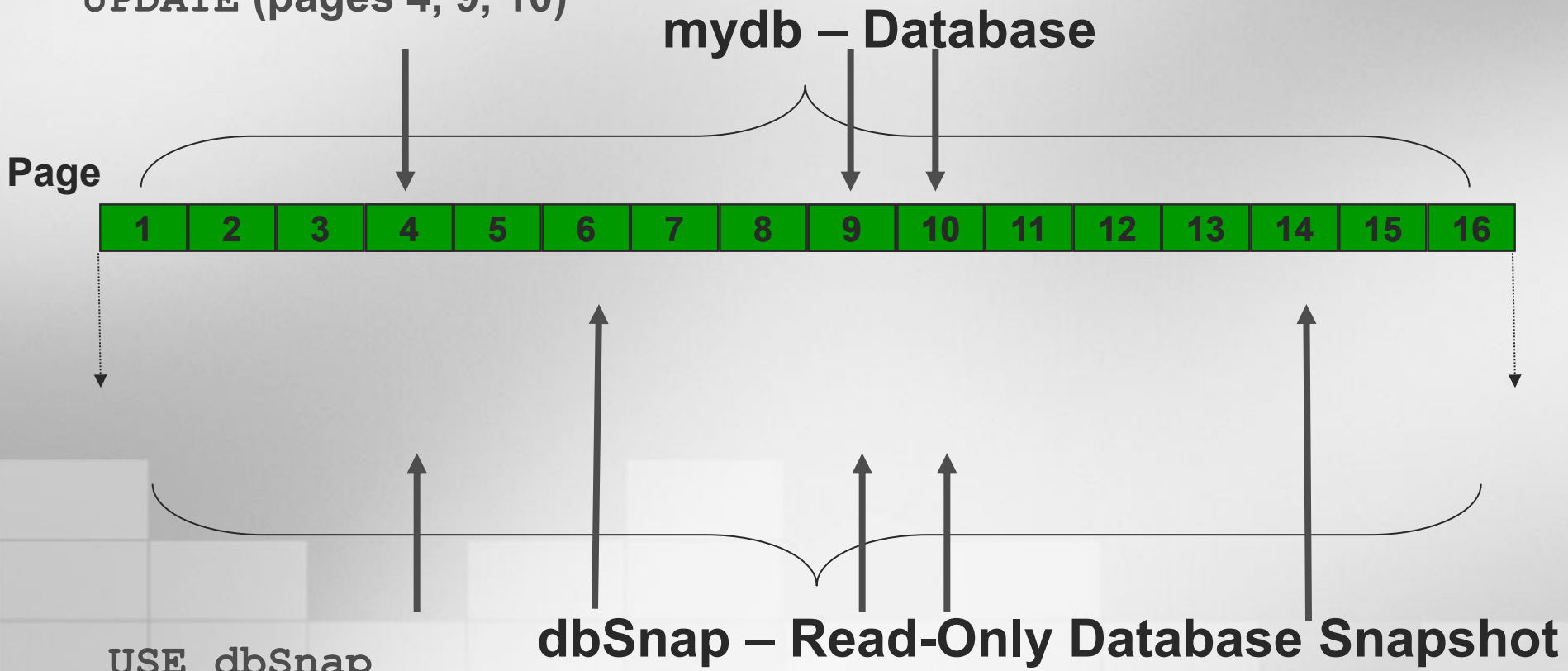



```
USE mydbSnap
SELECT (pages 4, 6, 9, 10, 14)
```

Database Snapshots

How does it work at the physical level

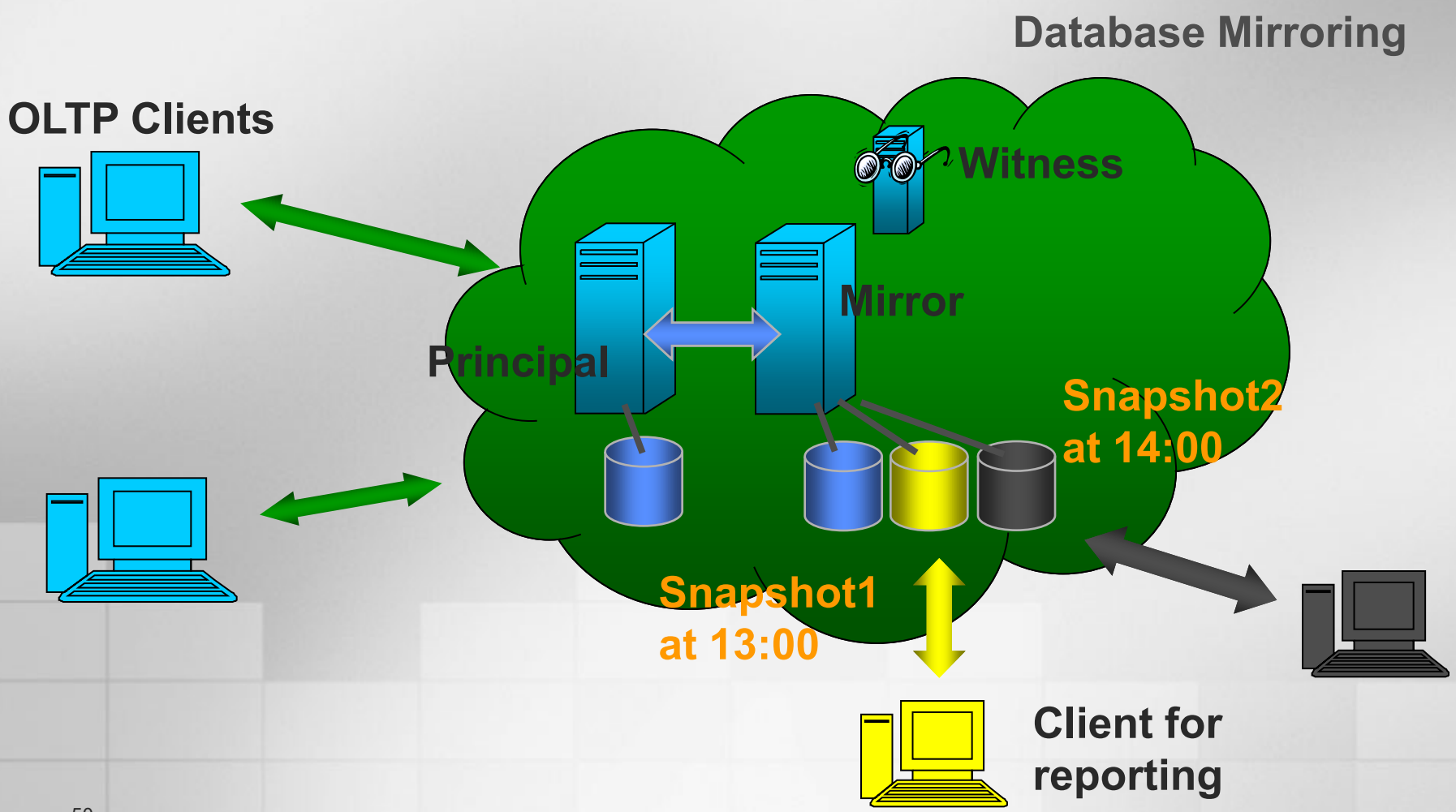
```
CREATE DATABASE dbSnap AS SNAPSHOT OF mydb
USE mydb
UPDATE (pages 4, 9, 10)
```



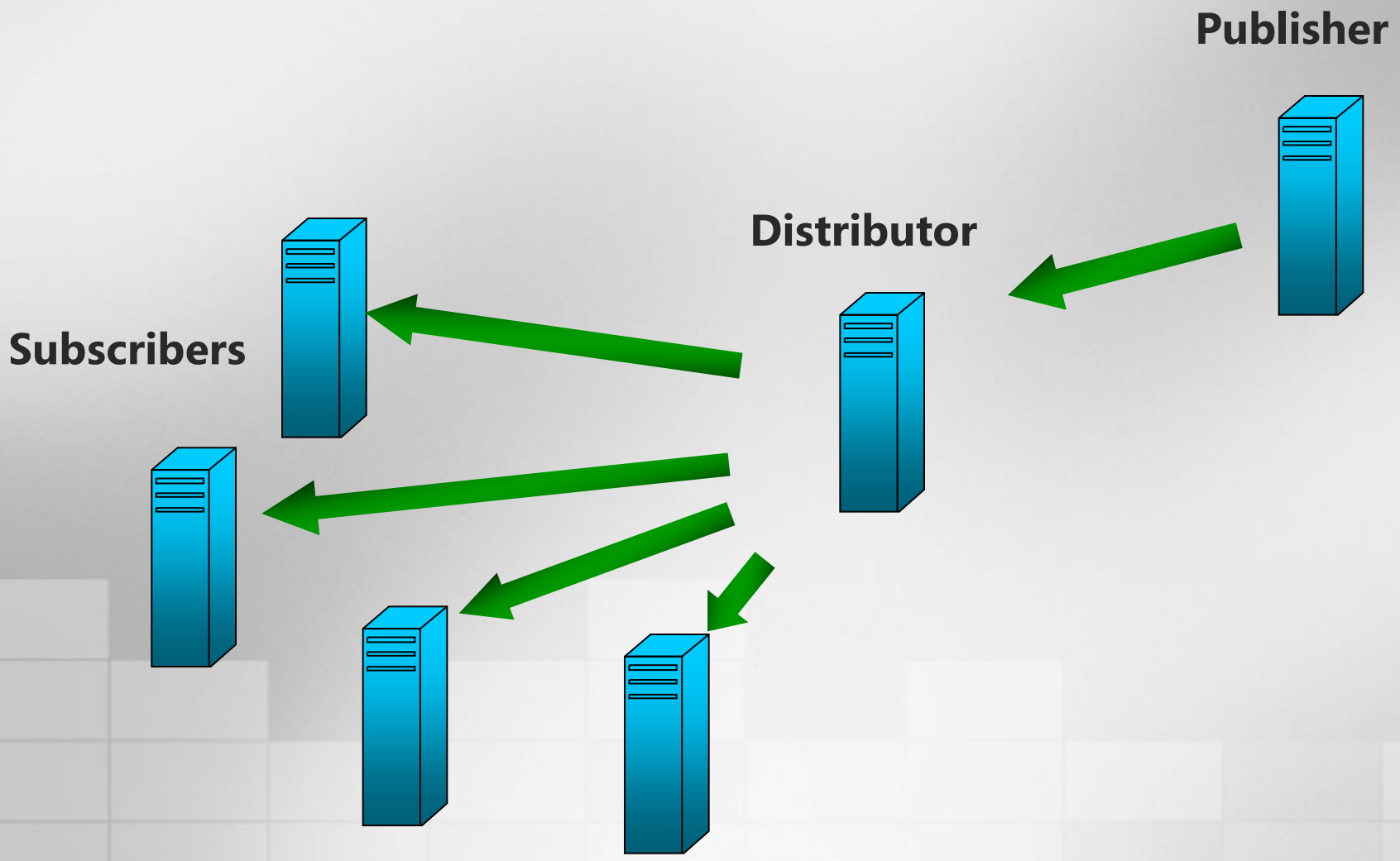

```
USE dbSnap
SELECT (pages 4, 6, 9, 10, 14)
```

Database Snapshots

Mirroring and reporting

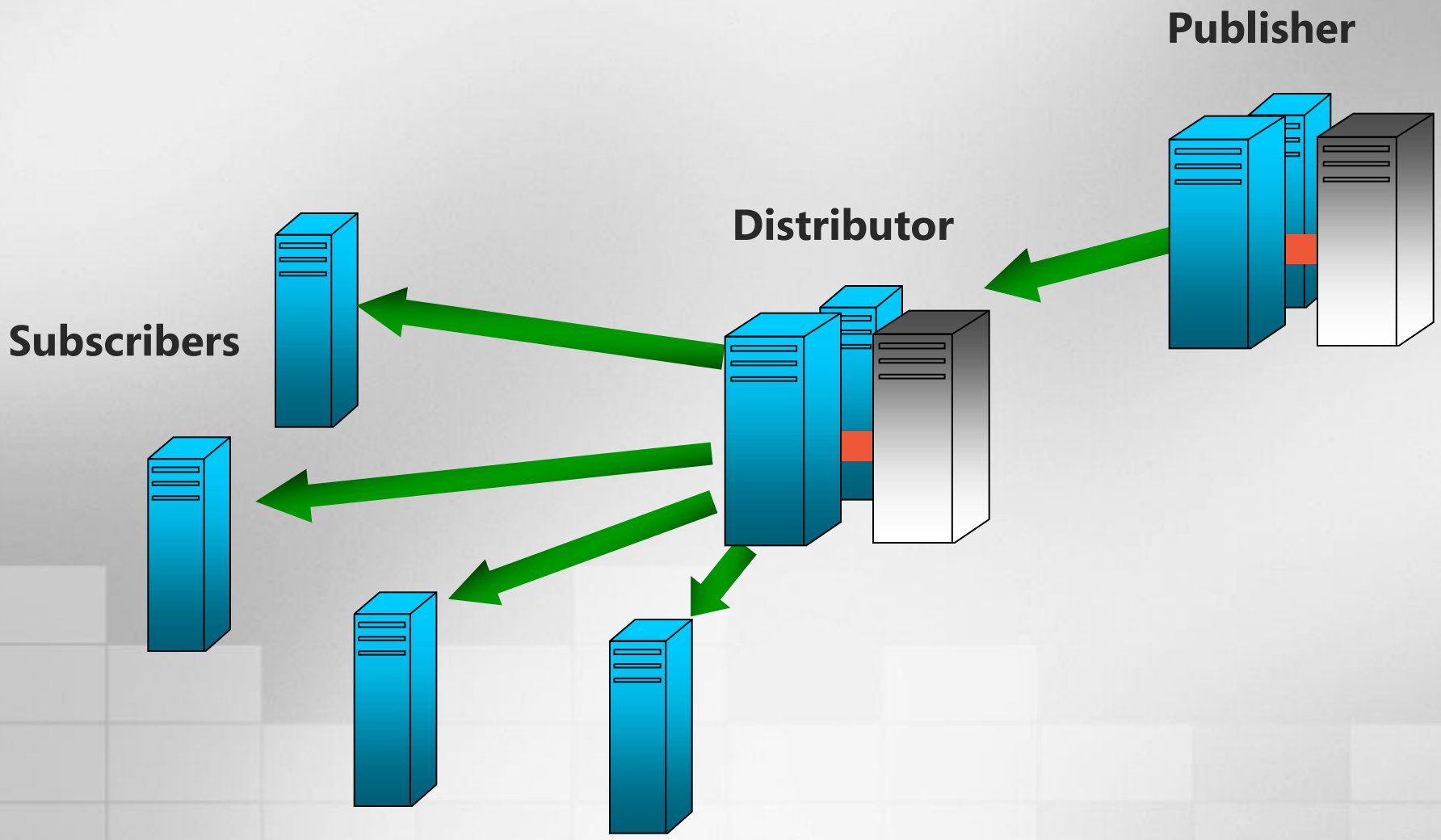


Replication
Overview



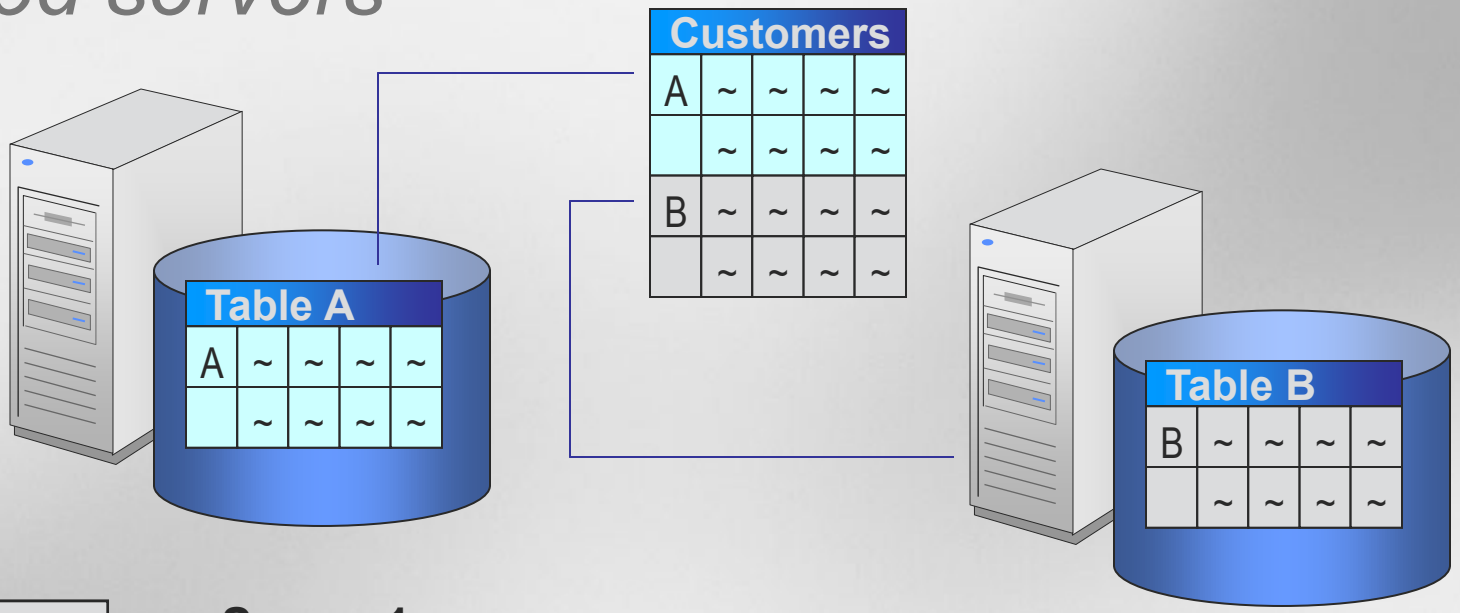
Replication and failover clustering

Complementary technologies



Shared Nothing Clustering

Federated servers

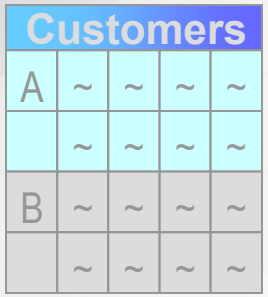


Server 1

Server 2

```
CREATE VIEW
  Customers
SELECT Table A
  UNION ALL
SELECT Table B
```

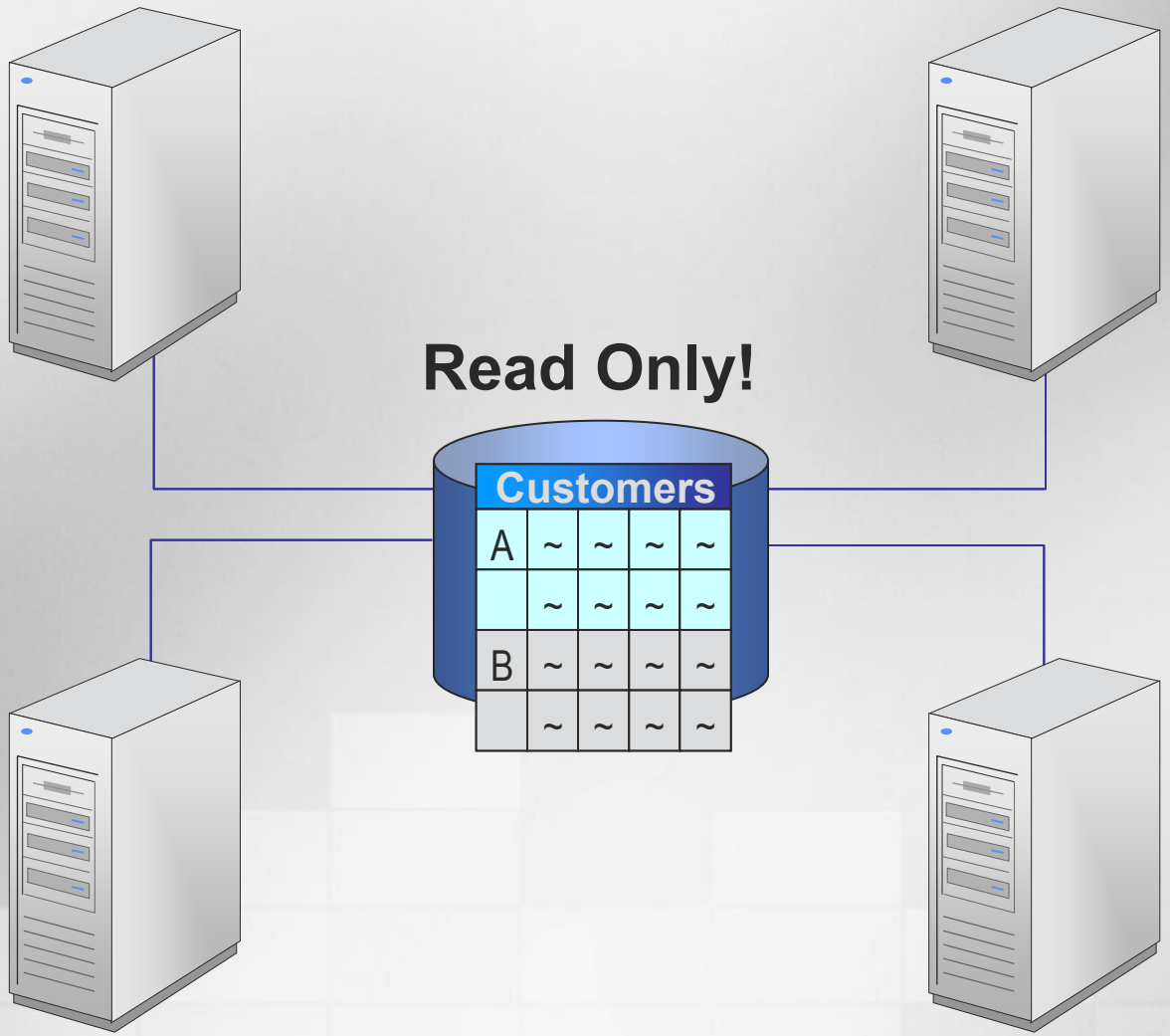
```
CREATE VIEW
  Customers
SELECT Table A
  UNION ALL
SELECT Table B
```



partitioned view

Shared Disk Clustering

Scalable Shared Databases



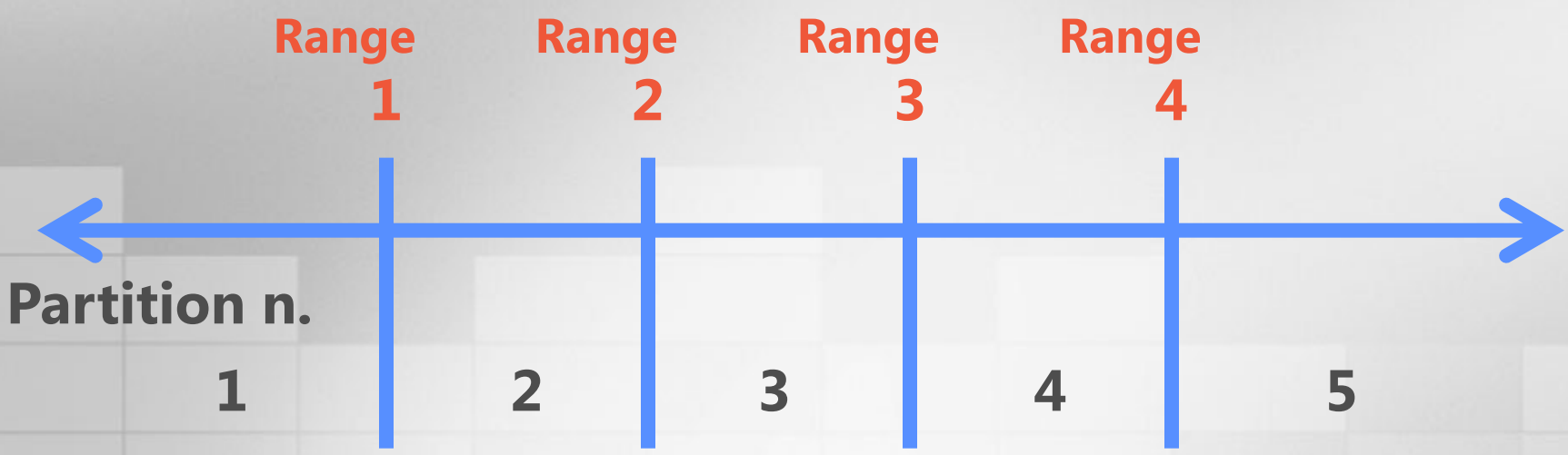
Shared Disk Clustering

Scalable Shared Databases

- More nodes can access the same database
- Read Only
- Not documented in original Books Online
  - first update December 2005
- Knowledge Base Article 910378
- Warning: no cache coordination between nodes

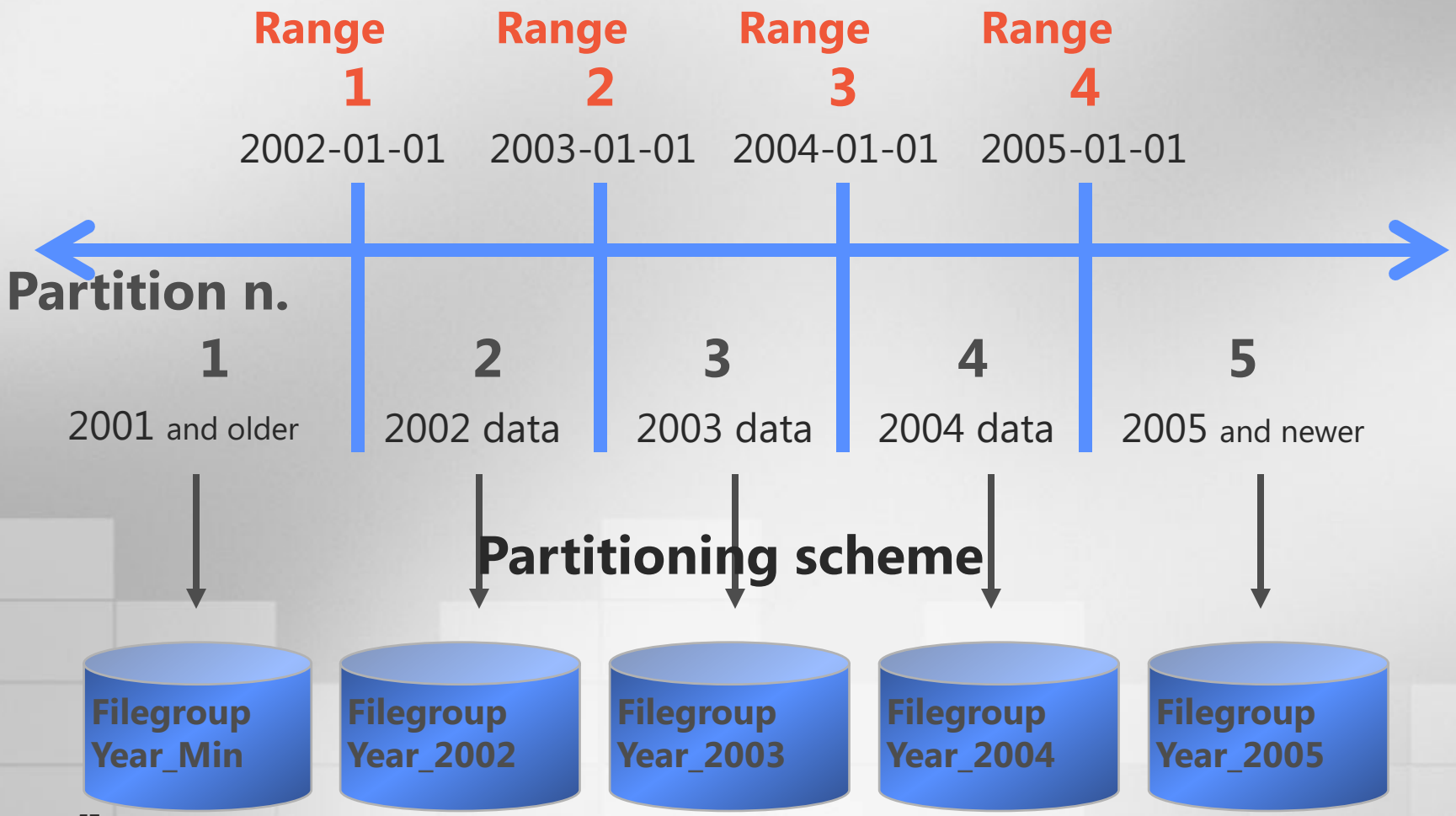
Data partitioning
Partitioning function

- Determines to which partition a specific row of a table or an index belongs
- Range partitioning
  - maps ranges of specific data types to integer values
  - defined using range values
  - N range values define N+1 partitions



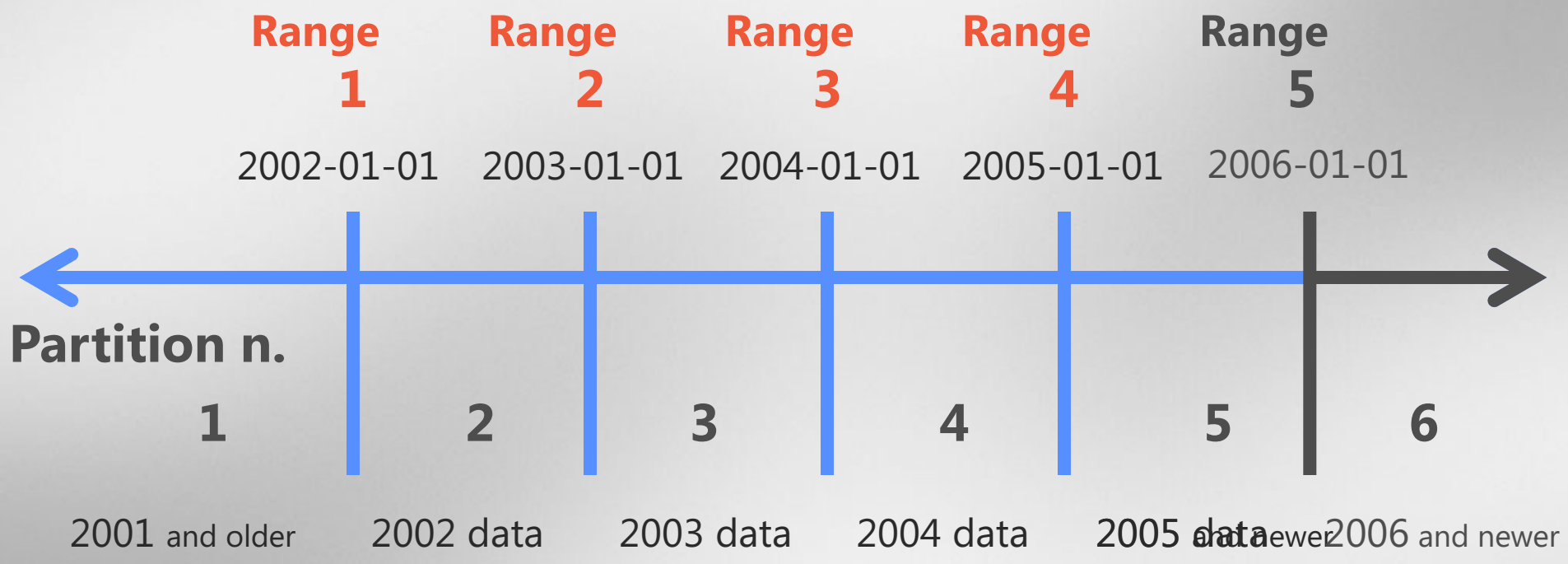
Data partitioning
Partitioning scheme

- determines where to store physically each partition of a partitioning function



Data partitioning

Split Range

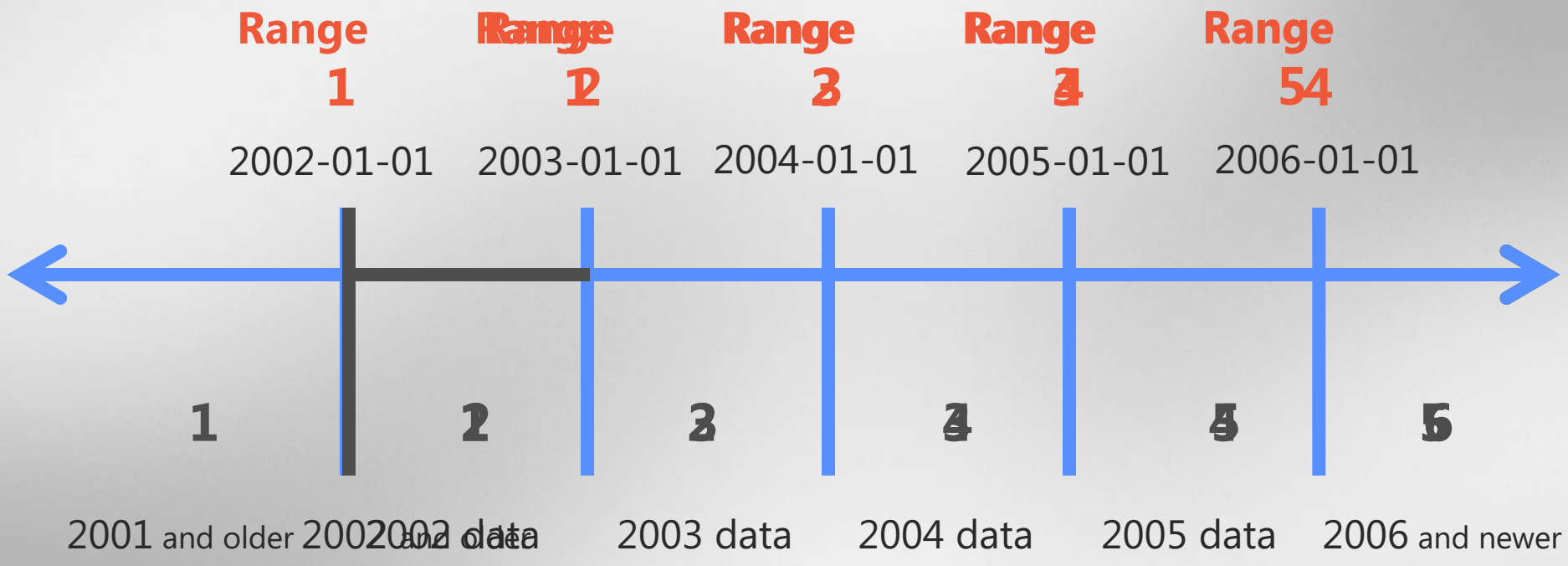



```
ALTER PARTITION FUNCTION annual_range()
SPLIT RANGE ('2006-01-01')
```

2006 data, and newer, moved over

Data partitioning

Merge Range

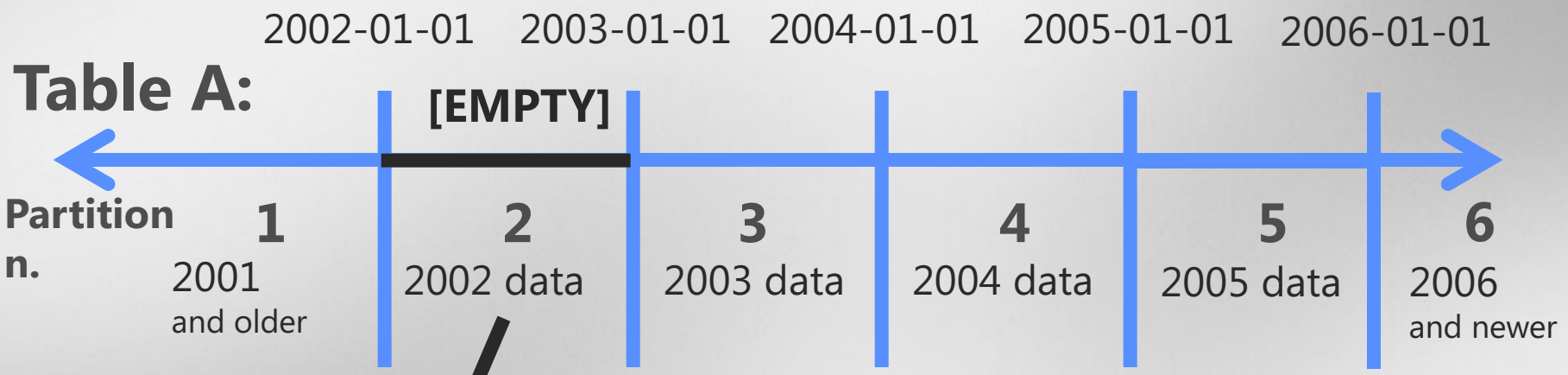


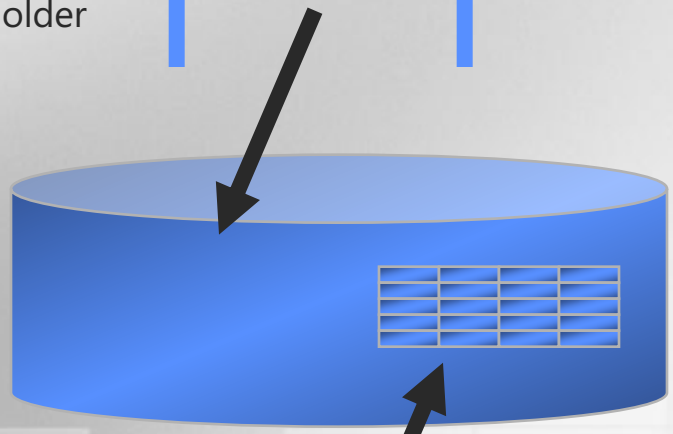
2002 data moved over

```
ALTER PARTITION FUNCTION annual_range()
MERGE RANGE ('2002-01-01')
```

Data partitioning

Switch Table To Partition





Filegroup DATA_2002

```
Alter Table B
SWITCH TO A
PARTITION 2
```

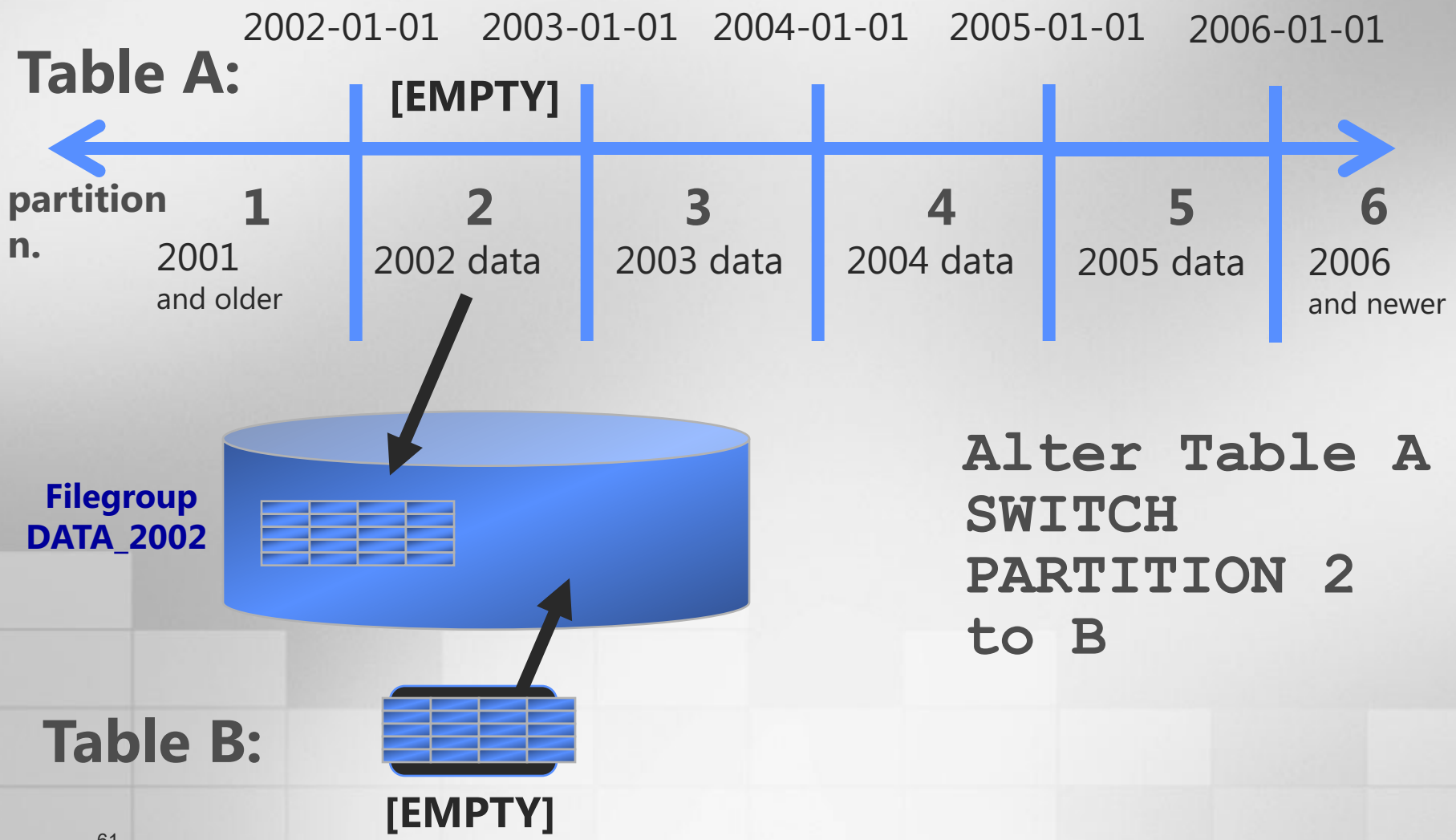
Table B:

[EMPTY]

CHECK CONSTRAINT:
B.Date_Key >= ‘2002-01-01’ and
B.Date_Key < ‘2003-01-01’

Data partitioning

Switch Partition To Table




```
Alter Table A
SWITCH
PARTITION 2
to B
```

Architectural Overview

Agenda

- Introduction
- Transaction Management
- High availability & scalability
- Security

Security model
Design Principles

- Secure by Design
  - Secure and robust code
  - Threat analysis and testing
- Secure by Default
  - Default configuration is a secure system
  - Discourage insecure configurations
  - Minimize attack surface
    - Install only necessary components
- Secure In Deployment
  - Principle of least privilege
    - Grant minimal permission required to function
    - Low privileged service accounts
  - Automate / Assist software maintenance
  - Good tools for security assessment / admin

2002 Focus
Emphasizing Secure Engineering

2003 Focus
Emphasizing Secure Default & Deployment

Security model

Security mechanism

- Authentication
  - Username and password
  - Certificates
- Authorization
  - Permissions
- Cryptography
  - Symmetric keys
  - Asymmetric keys

Authentication
Basics

- What is it?
  - Verify that someone is who they say they are
- Modes
  - Windows Integrated Logins (NTLM, Kerberos)
  - SQL Logins
- Consequence
  - SQL Server establishes a session with a security profile

Authentication

Login and password policy

- Windows Authentication
  - Windows users login
  - Windows password policy enforced
- SQL Server Authentication
  - SQL Server logins
  - Configuration of policies to enforce
    - complexity, expiration, change on next logon
  - Local machine or domain policies
  - Requires Windows 2003

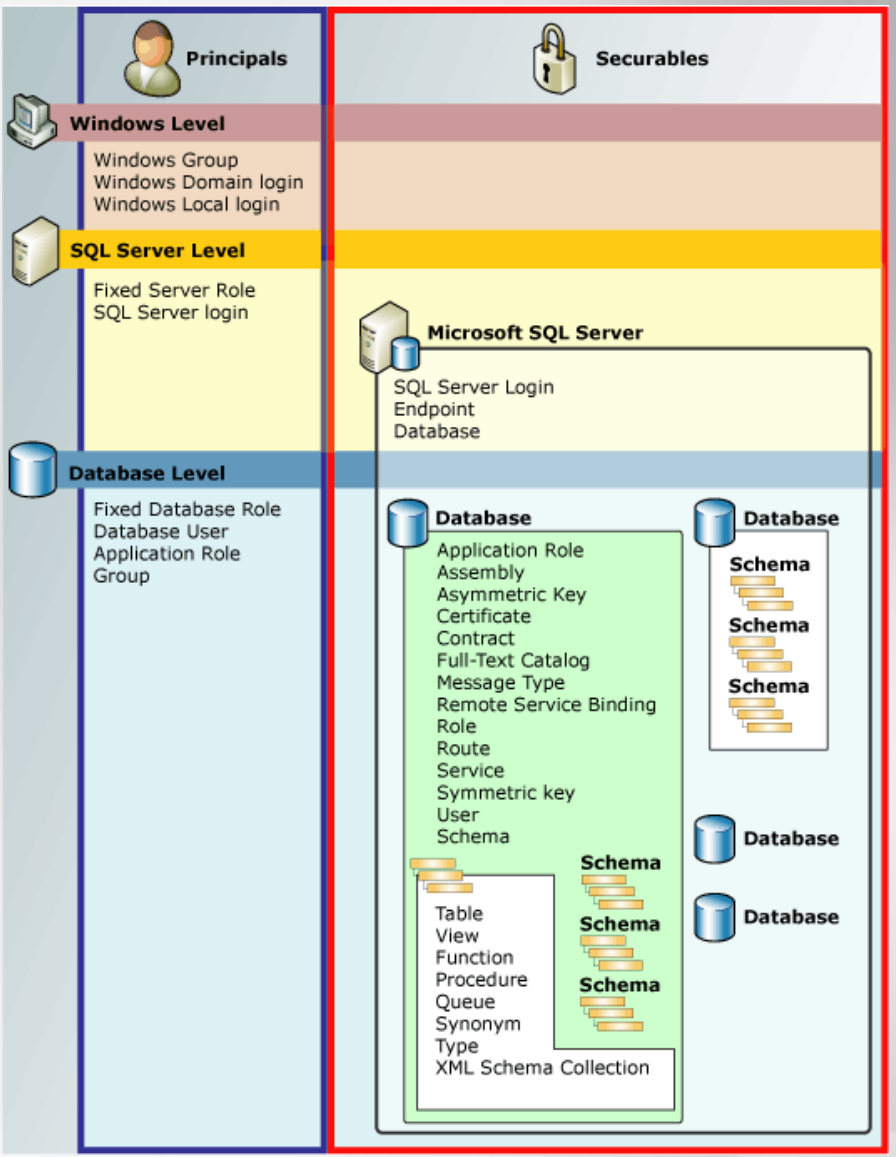
Authentication

Credential protection

- Login packets always encrypted
  - using SSL
    1. certificate configured for the instance
    2. certificate configured for the machine
    3. self-signed certificate generated by the engine
  - rest of communication not encrypted
    - can be if configured
  - use *self-signed* certificate only for authentication!

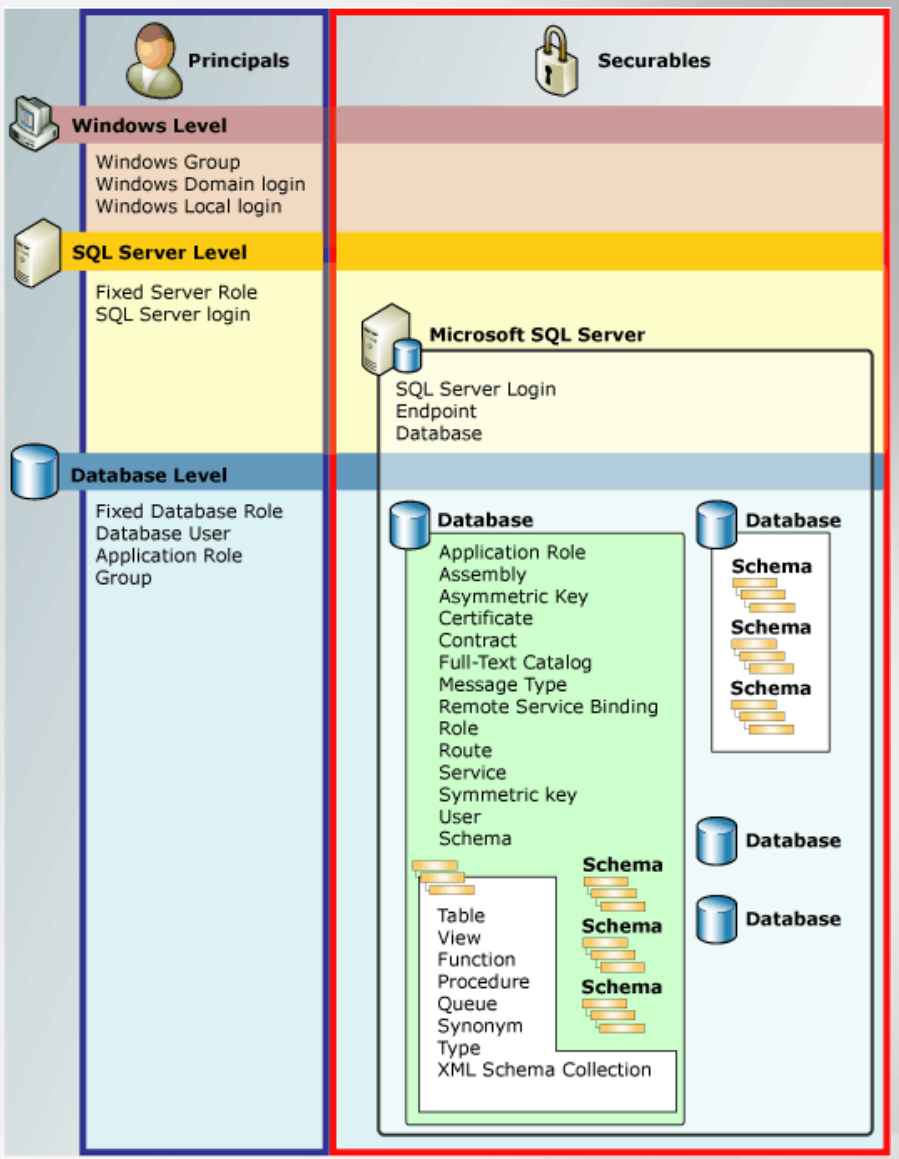
Permissions
Introduction

- Securables
  - Hierarchy of entities protected by specific permissions
- Principals
  - Subjects that interacts with securables according to specific permissions
- Permissions
  - Actions that can be done on securables
- Principle of least privilege



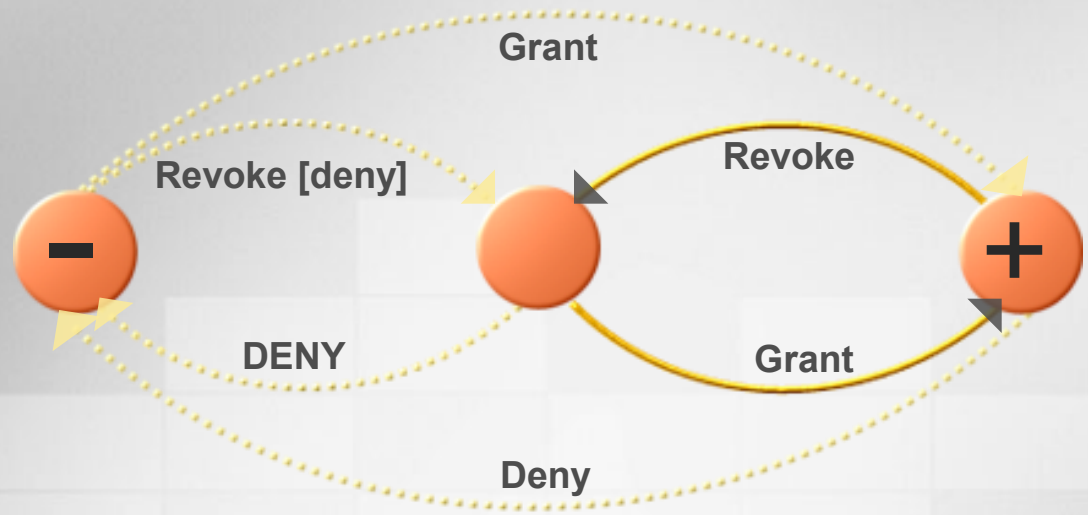
Permissions
Scope

- Server scope
  - Specific permissions for each securable
  - Fixed roles
- Database scope
  - Allow principals to execute tasks within the database
  - Custom roles
- Schema scope
  - Used to group database objects
- Inheritance of permissions



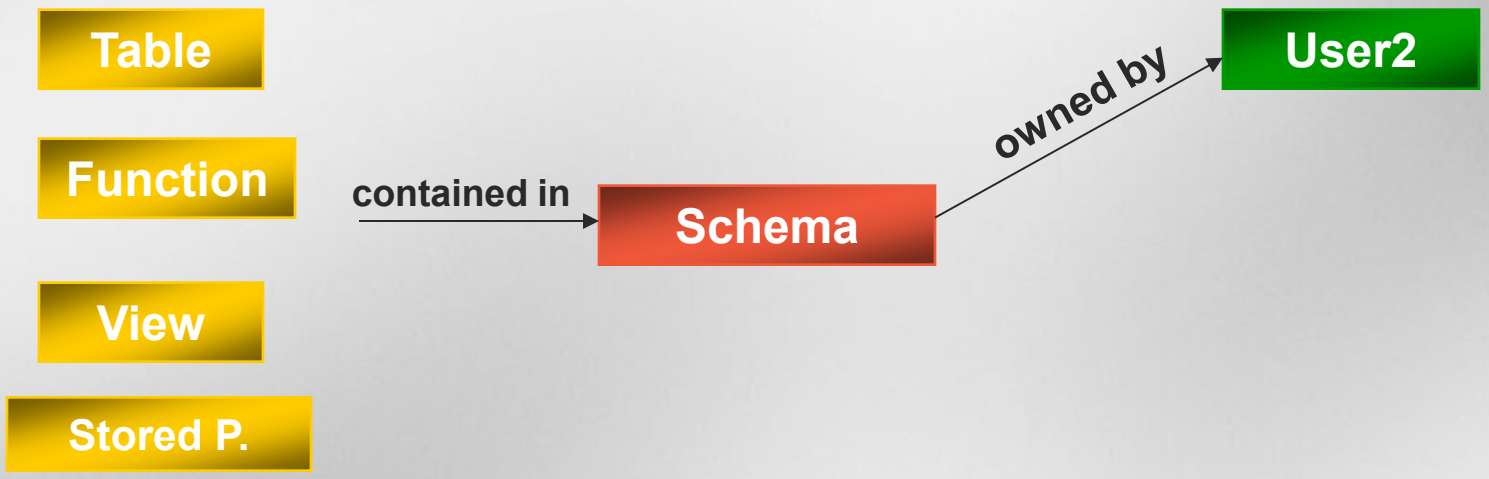
Permissions
Assigning

- GRANT
  - Gives a permission
- DENY
  - Forbids a permission
- REVOKE
  - Takes away a permission or a forbid

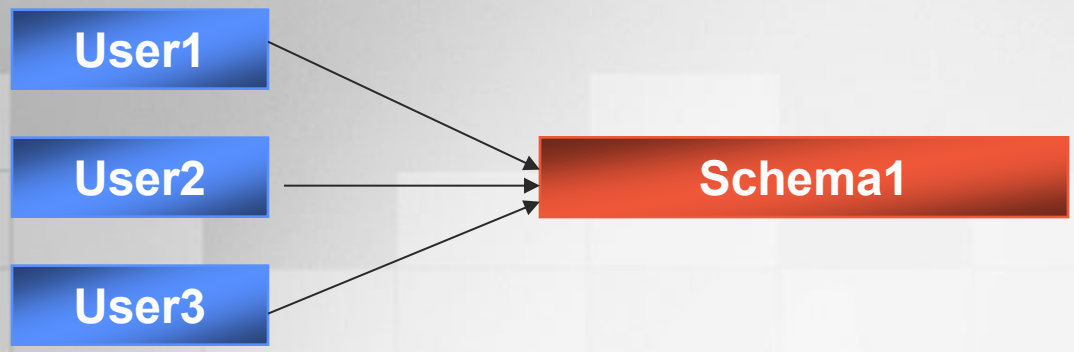


Users and schema

Overview



Default schema



Name resolution

```
SELECT * FROM Foo
```

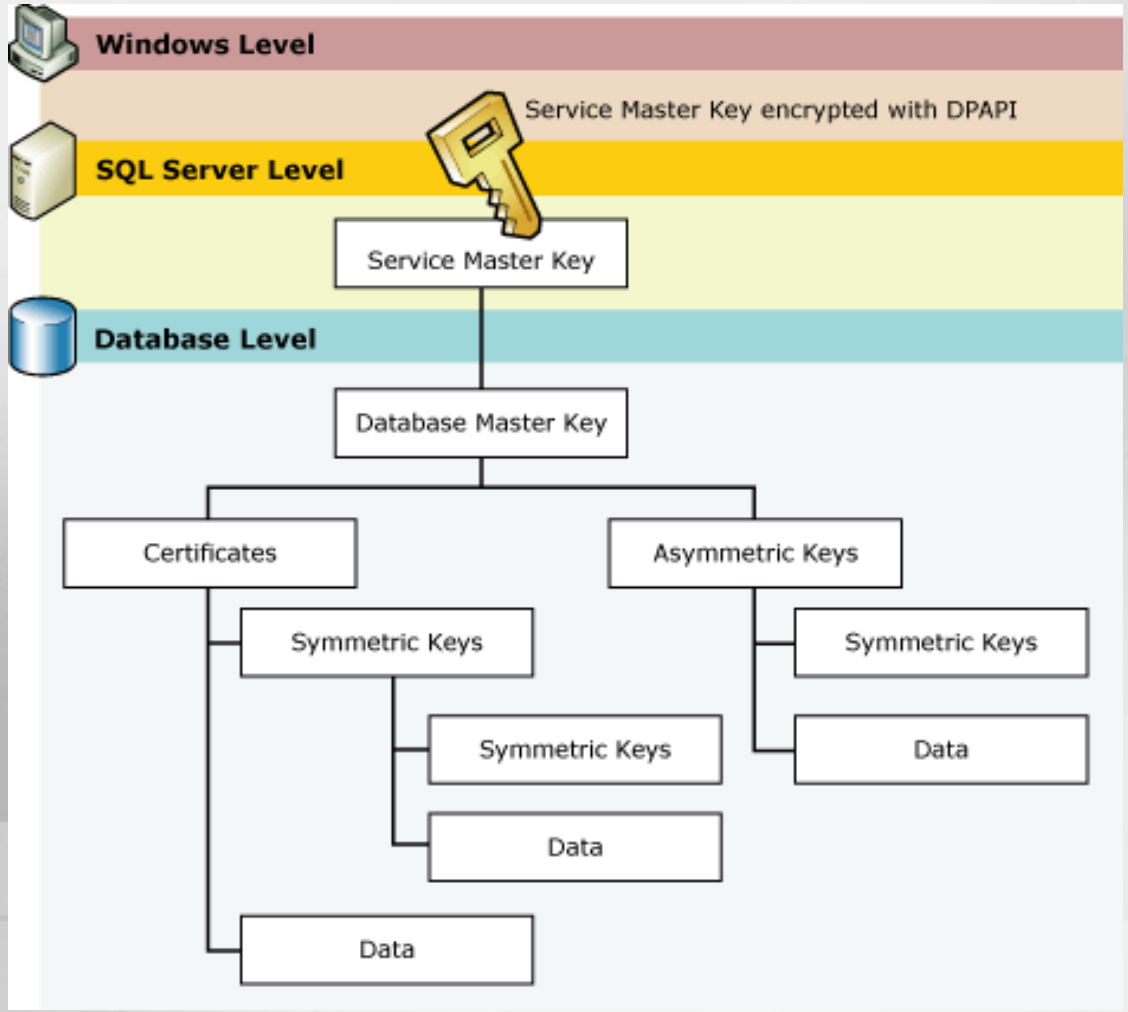
- Schema1.foo
- dbo.foo

Data Encryption
Introduction

- Complete infrastructure to manage
  - Symmetric keys
  - Asymmetric keys
  - Digital certificates
- Keys and certificates stored in the database
- Hierarchical key protection mechanism
- Association of keys and certificates to users

Data Encryption

Key protection hierarchy



Architectural Overview

Resources

- **Demo and presentation**
  http://solidqualitylearning.com/Blogs/gianluca
- **Information about the product**
  http://www.microsoft.com/sql
- **Support for IT professionals**
  http://http://www.microsoft.com/technet/prodtechnol/sql
- **Support for developers**
  http://msdn.microsoft.com/sql
- **SQL Server communities**
  http://www.ugiss.org
  http://www.microsoft.com/sql/community
  http://mvp.support.microsoft.com
- **Public Newsgroups via NTTP**
  microsoft.public.sqlserver.*
- **Pubblic Newsgroups via WEB**
  http://www.microsoft.com/technet/community/newsgroups/server/sql.mspx
- **Training and mentoring**
  http://www.microsoft.com/learning
  http://www.solidqualitylearning.com